FinOps Certified Practitioner (FinOps) Certification Crash Course

Prepare for the leading FinOps Certification

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Course Intro and Overview

Intro and Understanding the Course Content and Flow

Course Overview

SECTION 1:

CLOUD FUNDAMENTALS

SECTION 2:

FINOPS FUNDAMENTALS

SECTION 3:

FINOPS TEAMS AND MOTIVATION

SECTION 4:

FINOPS CAPABILITIES

SECTION 5 :
FINOPS LIFECYCLES

SECTION 6:
PRICING AND BILLING CONSIDERATIONS

SECTION 7:

EXAM PREPARATIONS

Learning Reinforcements

- Whiteboard Discussions
- Cloud Demos
- Test Tips
- Practice Questions eBook
- Practice Exams available on TechCommanders
- Community Sites on TechCommanders
- Documentation Links

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About Your Instructor

Joseph Holbrook, CLO of Techcommanders in Jacksonville, FL

- Certified Blockchain Solutions Architect (CBSA)
- Certified Google Cloud Platform Cloud Architect
- Certified AWS Solutions Architect
- FinOps Practitioner
- Brocade Distinguished Architect (BDA) 2013
- EMC Proven Professional – Expert – Cloud (EMCCE)
- Published Course Author on Pearson Safari, Udemy, LinkedIn Learning
- Published Book Author – Architecting Enterprise Blockchain Solutions
- CompTIA Subject Matter Expert, SME
- Prior US Navy Veteran

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Certification Overview

What is the FinOps Exam about and what will be tested on?

Objectives of the Exam

• Challenges of Cloud	8%
• FinOps & FinOps Principles	12%
• FinOps Teams & Motivation	12%
• FinOps Capabilities	28%
• FinOps Lifecycle	30%
• Terminology & the Cloud Bill	10%

Exam Overview

- Exam is proctored online by the Linux Foundation.
- 1 Hour to Take
- Certification is good for 3 years.
- Exam is 50 questions that are multiple choice
- You have one FREE retake if you do not pass first time

Course Pre-Requirements

What you need to be successful in the course.

Exam Pre-Requirements

- Understand the basics of how cloud computing works, know the key services on your cloud providers, including their common use cases, and have a basic understanding of billing and pricing models.
- Have a base level of knowledge of at least one of the three main public cloud providers
- Understand the value proposition of the cloud.

Course Overview

NO FINOPS PRE-REQUIREMENTS.

CLOUD EXPERIENCE

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Section 1 - Cloud Computing Fundamentals

Understanding what Cloud is all about

Section Overview

- INTRO TO CLOUD COMPUTING
- CLOUD COMPUTING CHARACTERISTICS
- CLOUD COMPUTING SERVICE MODELS
- CLOUD COMPUTING DEPLOYMENT MODELS
- COMPARING ON PREMISES TO CLOUD
- CHALLENGES OF CLOUD COMPUTING
- SECTION SUMMARY
- SECTION REVIEW QUESTIONS

Intro to Cloud Computing

What is the Cloud?

Defining Cloud Computing

SP 800-145 Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.

This cloud model is composed of five essential characteristics, three service models, and four deployment models.

NIST – SP-800-145

https://csrc.nist.gov/publications/detail/sp/800-145/final

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Cloud computing is the on-demand delivery of compute resources, database storage, applications, and other IT resources through a cloud services platform via the internet with pay as- you-go pricing

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Cloud or Virtualization?

- Virtualization is an enabling technology for cloud computing and cloud computing services. (Complementary)
- For cloud computing to occur, it is necessary to separate resources from their physical location.
- Without virtualization, the cloud becomes very difficult to manage.
- In addition, cloud computing is a business model where ownership of physical resources rests with one party, and the service users, which form the other party, are billed for their real use.

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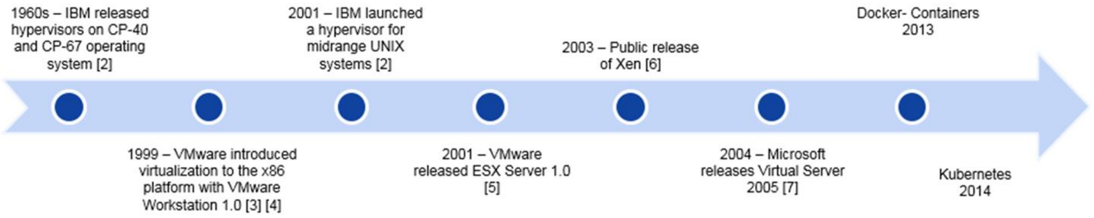

- Virtualization was used in mainframe computing as early as in 1960s by IBM.
- 1999 - VMware introduced virtualization into the x86 platform (PC environment).

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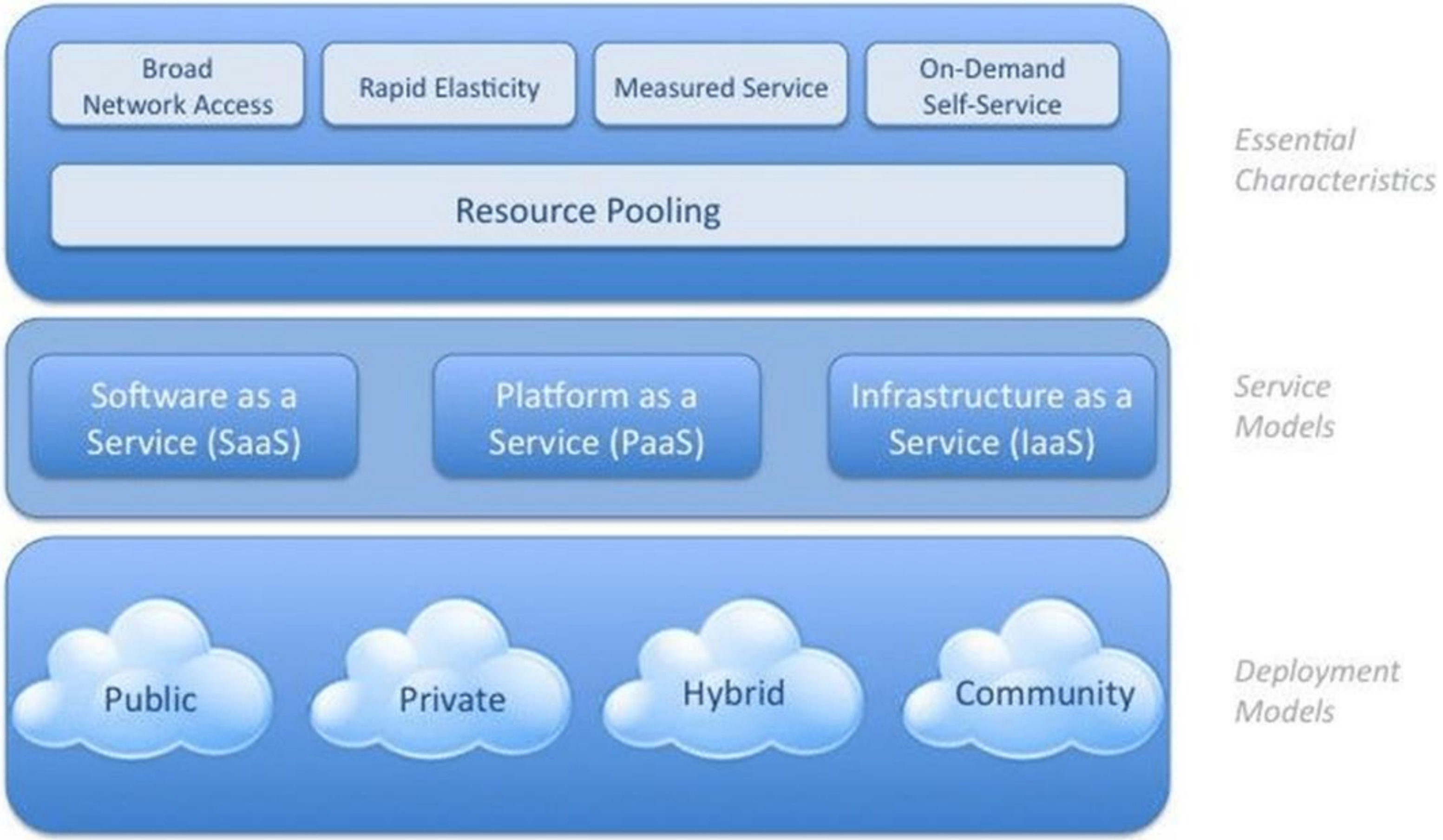


Why Use Cloud Computing?

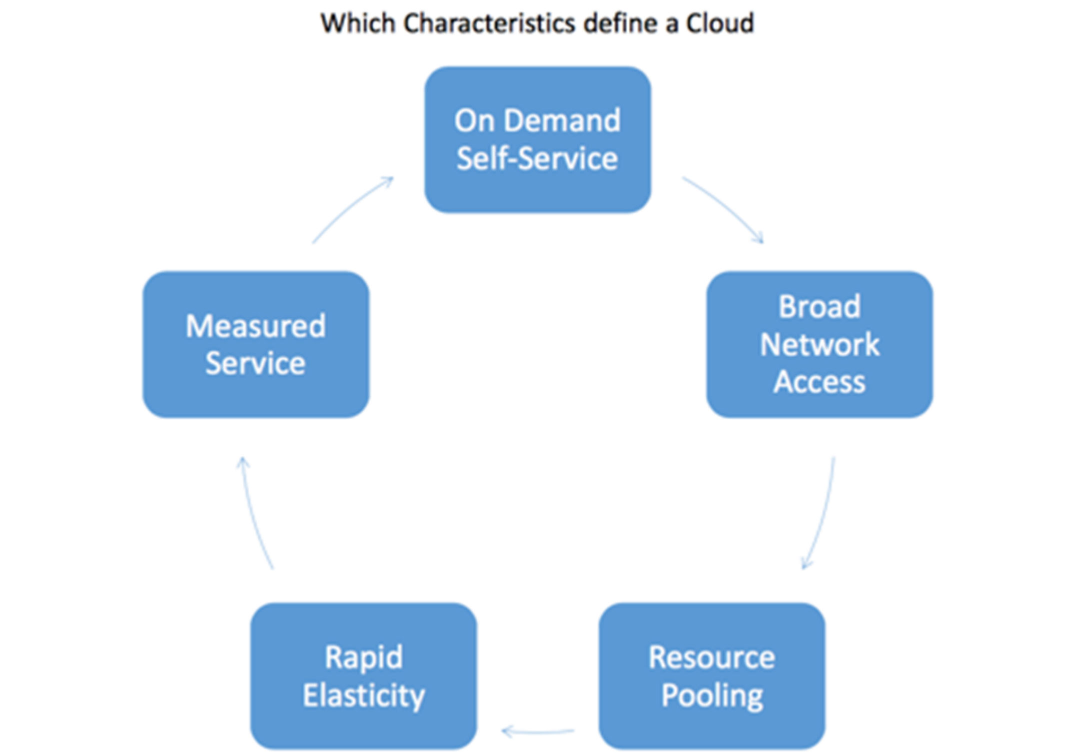
- Companies use cloud computing to leverage the resources of a cloud computing provider such as AWS, Azure and Google Cloud.
- Leverage enterprise grade data center resources such as data storage, analytics and compute services.
- Pay for what you use and not what do not need.
- On Demand and Flexible Model

Test Tips

- Cloud Computing is on demand delivery…..
- In Cloud Computing your pay for resources used.
- Resource consumption in cloud is variable.

Essential Cloud Computing Characteristics

What does it take to be a Cloud?



NIST Five Essential Cloud Characteristics

- On Demand Self Service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Measured Service

On-demand self-service

- A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

Broad network access.

- Capabilities are available over the network and accessed through standard mechanisms that promote use on any device.

Resource pooling.

- The provider’s computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.

Rapid elasticity

- Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.

Measured service

- Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service. Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

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Services	Description
On Demand Self Service	Resources are available as needed
Broad Network Access	Access anytime or anywhere (Internet)
Resource Pooling	Data Center resources are pooled together optimizing quality of service
Rapid Elasticity	Ability to add or remove computing resources based as needed
Measured Services	The ability to measure resources and charge customers

Test Tips

- Cloud Computing is on demand delivery…..
- Rapid elasticity allows for scaling based on consumer demand.
- Multi tenant models and virtualization allow for resource pooling

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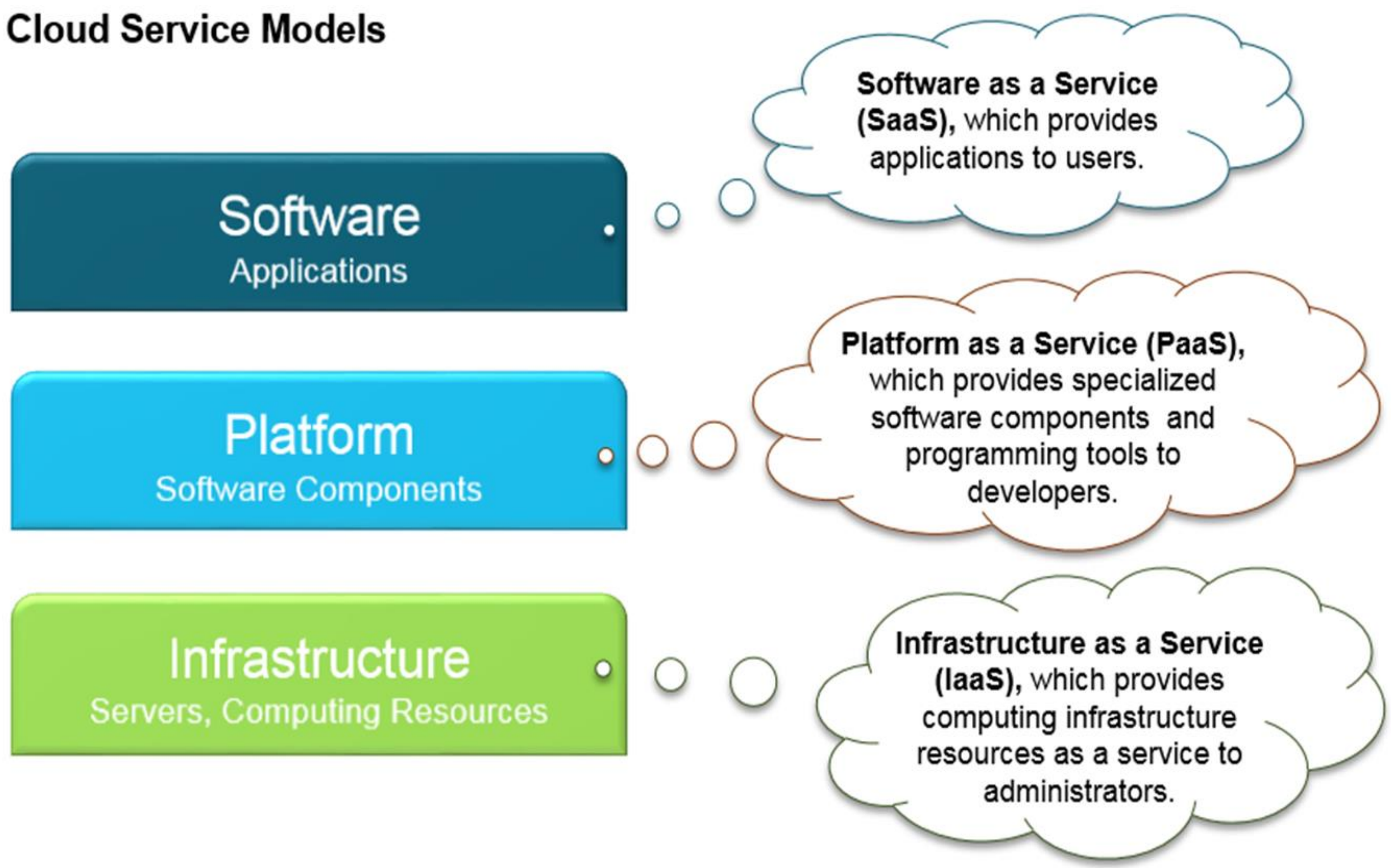
Cloud Service Models

What are the cloud service models?



Meet Business Requirements

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Infrastructure as a Service (IaaS) contains the basic building blocks for cloud IT and typically provides access to networking features, computers and data storage space.

IaaS provides the highest level of flexibility and management control over the infrastructure (Example – GCP Compute Engine and AWS EC2)

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Platform as a Service (PaaS) removes the need for your organization to manage the underlying infrastructure and allows you to focus on the deployment and mgmt. of your applications.

PaaS provides the second highest level of flexibility and management control over the infrastructure.   (Example – GCP App Engine and AWS Elastic BeanStalk)

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Software as a Service (SaaS) provides you a complete product that is run and managed by the service provider.

You worry only about using the software and not about infrastructure.

SaaS provides the lowest level of flexibility and management control over the infrastructure. (Example – Google Gsuite and MS O365)

Service	Description
IaaS	Data center resources: computer, network, and storage resources
PaaS	Platform resources: databases, web services, and other middleware applications
SaaS	Software resources: customer applications

Test Tips

- Three Service Models and Four Deployment Models
- SaaS provides the lowest level of flexibility while IaaS provides the highest level of flexibility.
- Billing is more complex to manage as we move to IaaS

Cloud Deployment Models

What are the cloud deployment models?

PRIVATE

HYBRID

COMMUNITY

PUBLIC

NIST Defines Four Cloud Deployment Models – How we use cloud

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Public – The cloud infrastructure is planned for the common public use. It may be possessed, handled, and functioned by some business enterprise, academic institution, or some government organization. (AWS, GCP, Azure)

Hybrid - The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability

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Private -The cloud infrastructure is planned for selective use by an individual organization which contains multiple customers. It may be possessed, handled, and functioned by the organization or a third party and it may hold on or off premises.

Community – The cloud infrastructure is planned for selective use by a particular community of consumers from organizations that have mutual interests like security needs, policy, and compliance considerations.

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On-premises – you run everything in your own DC

Hybrid – Combine two or more of the models.

Community – Shared deployment between organizations.

Public Cloud – you run all your services in a Public Cloud

Test Tips

- Three Service Models and Four Deployment Models
- Hybrid Model is used for bringing together two or more deployment models. (Public and Private)
- Private Cloud is used for enterprises that require high security, compliance and even performance.

Comparing On Premises to Cloud

On Premises vs Cloud

Some effects of moving from on premises to Cloud Computing:

- Localization of cost: From central management to a developer’s hands.
- Material spend: No possibility to predict the exact cost of the infrastructure.
- Variable price: Spend what you use.

Difference between Service Models?

- Mainly the functionality and services offered
- Shared Responsibility model (How much customer and provider perform)
- Like the Pizza Model

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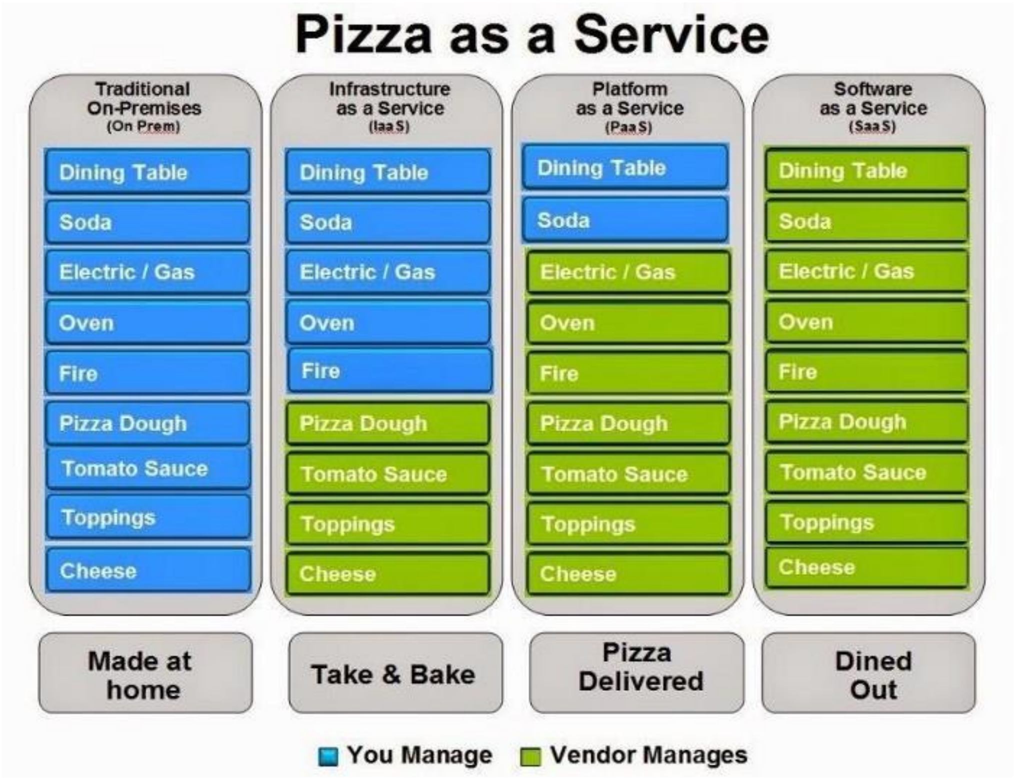
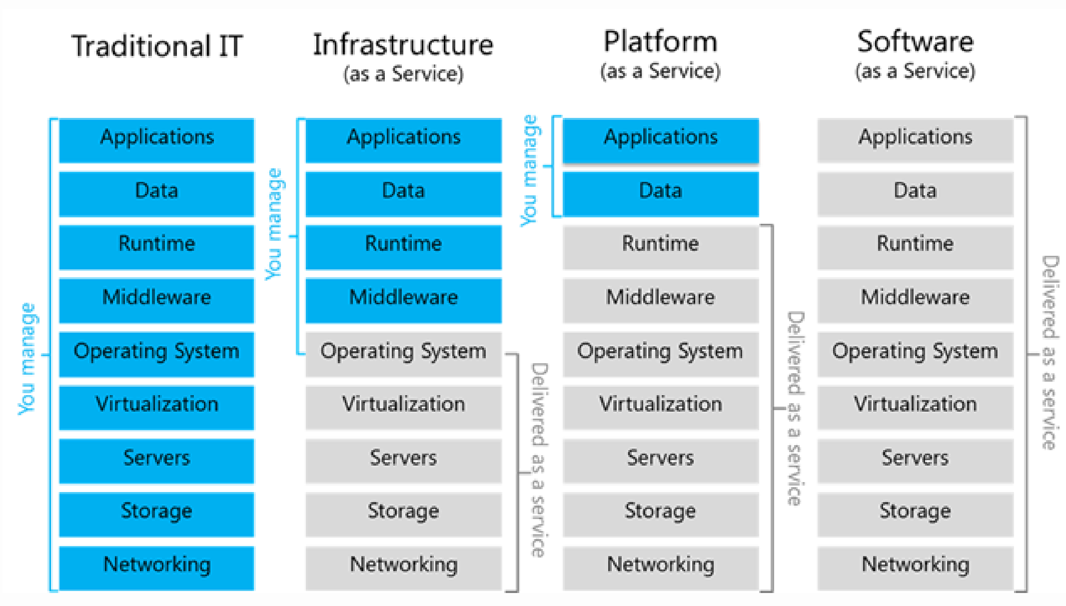


Graphic – Paul Kerrison



Graphic - VMware

Differences between on premises and cloud.

- Costs
- Resource Usage
- Performance
- Security
- Reliability
- Deployment
- IT Staff Requirements

Challenges of Cloud Computing

FinOps and Cloud bring new challenges

Challenges – Lets Discuss

Video

Section Summary

Section 1 : Cloud Computing Overview

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1

Section

Section Summary

- The NIST cloud model is composed of five essential characteristics, three service models, and four deployment models.
- Some common Cloud benefits are Low-Cost Services, Agility and Elasticity, Open and Flexible, Secure.
- NIST Five Essential Cloud Characteristics are on-demand self-service, broad network access, resource pooling, rapid elasticity or expansion, and measured service.
- Hybrid Cloud is when we combine our on-premises apps with GCP.
- There are three Cloud Service Models – IaaS, PaaS, SaaS
- There are four Cloud Deployment Models – Private, Public, Hybrid and Community
- Challenges with Cloud for FinOps will ranges from VM templates, APIs, Network usage and other services tied to the VM and application.
- Cloud is using a variable spend model which is very different from on premises.

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Section Review Questions

Section 1 : General Cloud Overview

Review Questions

Which of the following NIST Five Essential Cloud Characteristics would be used for enabling the sharing of resources between cloud tenants? (Select One)

- On Demand Self Service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Measured Service

Review Questions

Which of the following NIST Five Essential Cloud Characteristics would be used for enabling the sharing of resources between cloud tenants? (Select One)

- On Demand Self Service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Measured Service

Review Questions

Cloud uses a variable spend model? True or False (Select One)

- True
- False

Review Questions

Cloud uses a variable spend model? True or False (Select One)

- True
- False

Review Questions

Which of the following service models provides software to users? (Select One)

- IaaS
- PaaS
- SaaS
- DaaS

Review Questions

Which of the following service models provides software to users? (Select One)

- IaaS
- PaaS
- SaaS
- DaaS

Review Questions

Which of the following service models provides a development environment for users? (Select One)

- IaaS
- PaaS
- SaaS
- DaaS

Review Questions

Which of the following service models provides a development environment for users? (Select One)

- IaaS
- PaaS
- SaaS
- DaaS

Review Questions

Which of the following would NOT be considered a characteristic of cloud computing? (Select One)

- On Demand Self Service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Simple Integration
- Measured Service

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Review Questions

Which of the following would NOT be considered a characteristic of cloud computing? (Select One)

- On Demand Self Service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Simple Integration
- Measured Service

Section 2 – FinOps Fundamentals

What is FinOps and the basics.

Section Overview

WHAT IS FINOPS	Why Use Finops	FinOps History	FinOps Six Principles
FinOps Lifecycle	CAPEX and OPEX	ROI and TCO Fundamentals	Chargeback
Cost Optimization Challenges	FinOps Terminology	Section Summary	Section Review Questions

What is FinOps

Let's Define and Identify what this really means.

What is FinOps about?

- The practice of bringing financial accountability to the variable spend model of cloud, enabling teams to make business trade-offs between spend, cost and quality.

  -FinOps Foundation

What is FinOps about?

- FinOps is a collaborative effort between Finance and Operations where cloud spending is moderated and optimized.
- AKA – Cloud Financial Management
- A culture to adopt
- Method of ROI and cost optimization

What is FinOps about?

- An efficient approach for teams to manage their cloud costs.
- Ownership driven and supported by a central group
- Work towards financial and operational control.
- A way to prioritize ongoing optimization of your cloud costs.

Test Tips

- FinOps is about culture in your enterprise. The culture of collaboration needs to be there for success.
- FinOps is meant to help your enterprise to prioritize ongoing optimization of your cloud costs and reduce waste.
- Provide a better ROI. (Profit)

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Why use a FinOps Approach

Does Cloud Really need FinOps?

Why FinOps?

- Cloud Spending increasing YoY
- Managing Cloud Spending can be overwhelming w/o a culture shift
- Set of Best Practices
- Provide corporate oversight of cloud spending. (Centralized)

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FinOps History

How did we get here?

FinOps History

- Unofficially practiced in early cloud days as Cloud Cost Optimization
- FinOps is a result of cloud computing procurement being centralized.
- Business Units vs Finance Professionals make decisions
- FinOps Foundation - 2019

Principles of FinOps

Six Principles to Understand

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Six Principles of FinOps

1. Collaboration
2. Business Value
3. Ownership
4. Timely Reports
5. Centralized Team
6. Variable Cost Model

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FinOps is a process and a set of best practices to bring the following aspects to companies and stakeholders:

- Collaboration – FinOps is not about one person at a company but a constant collaboration between engineers and their managers, between R&D, Operations and Financial departments, CTO, CIO and VPs offices.
- Business Value – FinOps needs to be implemented to provide value for the organization. Provide for cost optimization and a solid ROI
- Ownership - Stakeholders need to be able to control and take ownership of their expenses.
- Timely Reports – Reports should be routine and concise to provide insight into cloud spend and forecasting.
- Centralized Team - A FinOps team should be centralized to provide for an enterprise focused approach removing redundancy but provide for collaboration.
- Variable Cost Model – Cloud Spending is variable and therefore the FinOps team must have processes, practices and tools in place to manage and monitor spending to meet the business goals.

FinOps Lifecycle

There's a documented journey with FinOps

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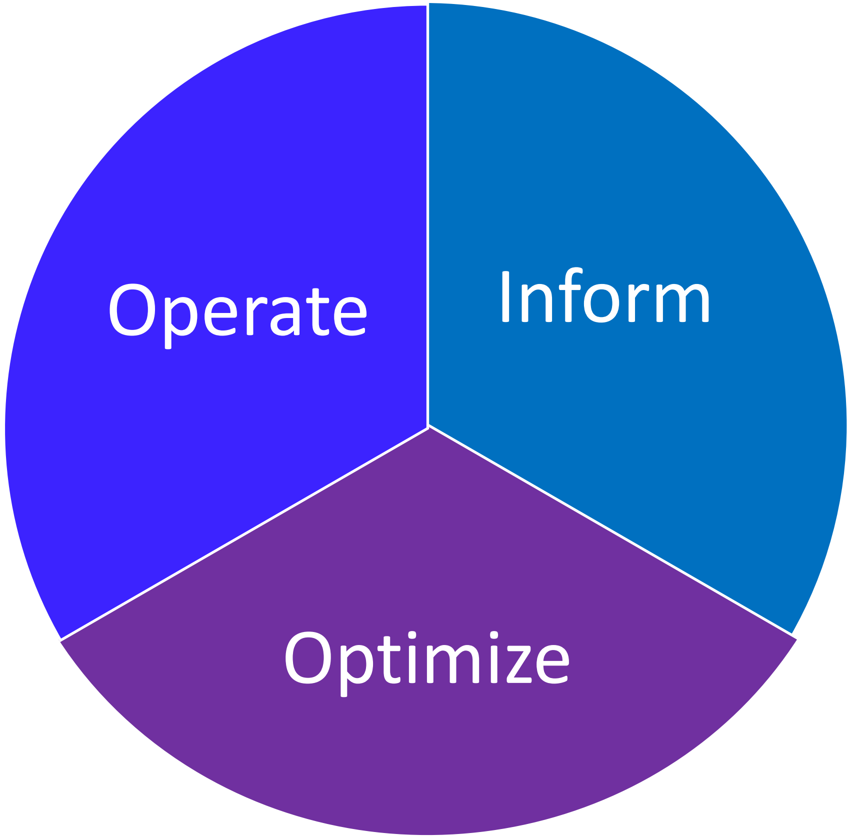
What is the FinOps Lifecyle?

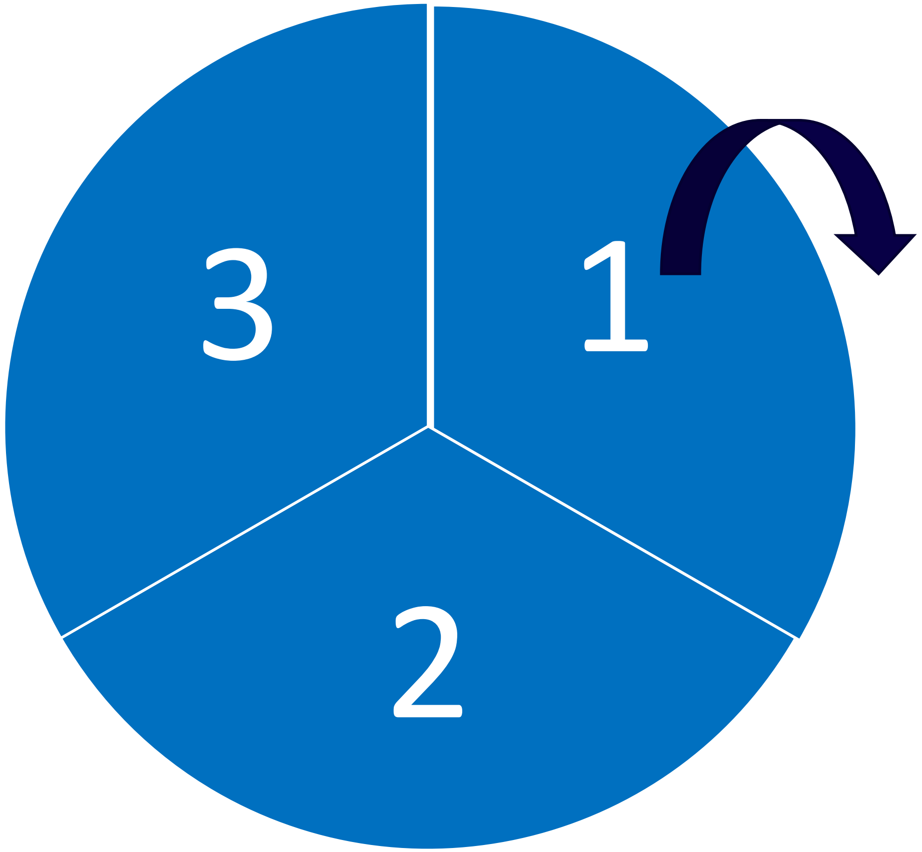
“The FinOps Lifecycle surrounds the idea of supporting a Culture of Accountability and a Governance framework to support the decision making FinOps makes possible”

- Inform
- Optimize
- Operate

Phases of the FinOps Lifecycle



Phases of the FinOps Lifecycle



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INFORM

- Understand your organizations resource consumption thru context, data, budgets and Forecasting.(Visibility and Allocation)

OPTIMIZE

- Setting your goals and targets by determining you cloud costs thru strategy, tracking, cost breakdown and cost growth boundary.(Utilization)

OPERATE

- Aligning organization teams to business goals thru drive and action. Focus is on getting results for the organization. (CI)

Lifecycle Phase	Main Tasks/Goals	Principles Supported
Inform	- Create tags/labels - Setup account hierarchy - Goal is for near real time cost visibility	Set Goals
Optimize	- Set the goals and determine the impact. - Achieve the goals	Principle 1: Teams need to collaborate; Principle 2: Business value of cloud drives decisions; Principle 6: Take advantage of the variable cost model of the cloud
Operate	- Target, define, and document optimization opportunities - Automation	Principle 2: Business value of cloud drives decisions Principle 5: A centralized team drives FinOps Principle 6: Take advantage of the variable cost model of the cloud

Test Tips

- Three Phases
- Create goals in the inform phase and then build processes in the operate phase to achieve goals
- Inform Phase – Visibility and Allocation
- Optimize – Utilization
- Operate - CI

Whiteboard - Considerations

Discuss Lifecycle

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Putting it all Together

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Test Tips

- Understand that Cloud is shifting spending from Capex to a Opex funding model.
- Cloud is generally paid for as a subscription using Opex funding.

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CAPEX and OPEX

What's the difference

CAPEX and OPEX

- Cloud computing introduces new ways to consume and pay for IT services.
- The shift from CAPEX to OPEX financing and cash flow models using cloud

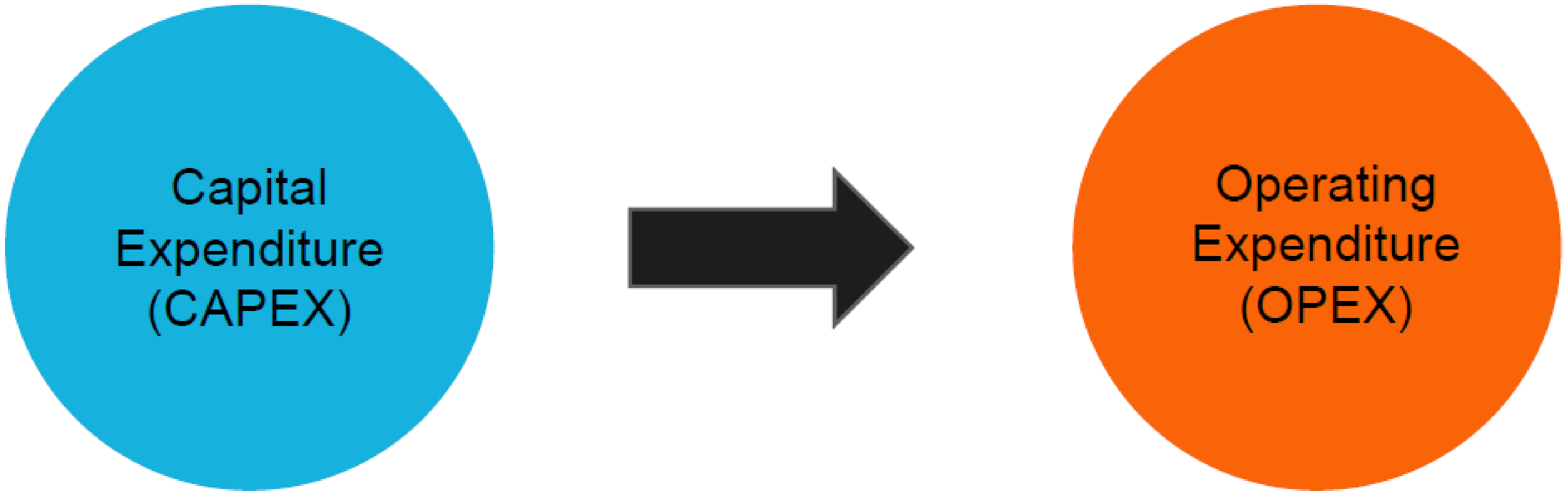
CAPEX and OPEX

- Capital Expenditures (Capex) are investments made by an organization for long-term benefits in the future.
- Operational Expenditures (Opex) are the ongoing costs related to day-to-day operations

	Capital Expense (CapEx)	Operating Expense (OpEx)
Purpose	Assets purchased with a useful life beyond the current year	Ongoing costs to run a business
When Paid	Lump sum up front	Monthly or annual recurring
When Accounted For	Over a 3 to 10 year lifespan as the asset depreciates	In the current month or year
Listed As	Property or equipment	Operating cost
Tax Treatment	Deducted over time as asset depreciates	Deducted in the current tax year
Example	Buying a laser printer	Buying toner for printer

CAPEX and OPEX

- Cloud computing is a shift from CAPEX to OPEX financing and payment models. The method of financing IT is changing with cloud computing.
- CAPEX is about ownership of assets, whereas OPEX is about paying for services.



Test Tips

- Understand that Cloud is shifting spending from Capex to a Opex funding model.
- Cloud is generally paid for as a subscription using Opex funding.

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ROI and TCO

Understanding ROI factors

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- Return on Investment (ROI) is perhaps the most widely used measure of financial success in business.
- ROI is the proportionate increase in the value of an investment over a period.
- Return on investment is typically calculated by taking the actual or estimated income from a project and subtracting the actual or estimated costs.
- That number is the total profit that a project has generated or is expected to generate.
- Lastly, this number is then divided by the costs.

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What is TCO

- Cloud TCO (total cost of ownership) is the total costs associated with your new cloud technology which is often compared to the total costs of your former server or data center deployment.
- The consumer's Total Cost of Ownership (TCO) of IT services changes while shifting to cloud computing.
- Consumers aim to reduce and optimize the TCO expenditure and increase value to business by leveraging new cloud services.
- Consumer IT service transition to new cloud service typically increases in transition and then falls as incremental OPEX model is adopted.

TCO vs ROI

An ROI calculation quantifies both the costs and the expected benefits of a specific project over a specific timeframe. (3-5 years)

ROI has both tangible and intangible components

TCO, on the other hand, includes just costs.

Goal is obtaining a solid ROI and LOW TCO

Benefits or ROI Measurements

- Reduced or eliminated software license costs
- Increased business agility and faster responses to business unit inquiries
- Reduced operational expenses – no more servers or ancillary infrastructure
- Shared cloud resources that all user departments can benefit from
- Improved energy efficiency and reduced electricity bills
- Retiring of IT assets, racks, data centers, and real estate expenditures

Measure Cloud ROI

- Optimization
- Usage
- Compliance Improvement
- Capacity Improvement
- Reduction in labor costs
- Faster TTM

Measure Cloud ROI

- ROI = net benefits/total cost
- (Gain from investment – investment) / (investment) = ROI
- Cloud Computing, we want to use Key Performance Indicators(KPI)
- KPIS such as capacity and utilization.

Test Tips

- ROI is used to measure the success of a potential or ongoing project.
- Goal is obtaining a solid ROI and LOW TCO

Chargeback

Understanding resources usage

What is Chargeback?

- Chargeback is a policy of allocating resource usage costs to actual business units.
- Improves Cloud Spending
- Allocates responsibility to departments, groups, locations, etc.
- Accounting Model

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Characteristics of Chargeback

IT chargeback accounting model, individual cost centers are charged for their IT service based on use and activity. (Uses tagging)

As a result, all IT costs are “zeroed out” because they have all been assigned to user groups. ...

Chargeback allows users/business units to see their costs and understand how those costs are determined.

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Test Tips

- Chargeback is a common method for organizations to mitigate overspending.
- Chargeback provides visibility and accountability for resource spending
- Accounting Model
- Tagging implementation

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Cost Optimization Challenges with DevOps

Understanding the dynamics to cost control with DevOps

Optimization Challenges

- DevOps consumes resources faster and unpredictably
- Cost is not always a concern with DevOps engineers. Cost vs Performance
- Hard Coded Resources (IaC)
- App Owner Resistance
- Traditional FinOps cost allocation doesn't work.
- Multiple Container/Apps

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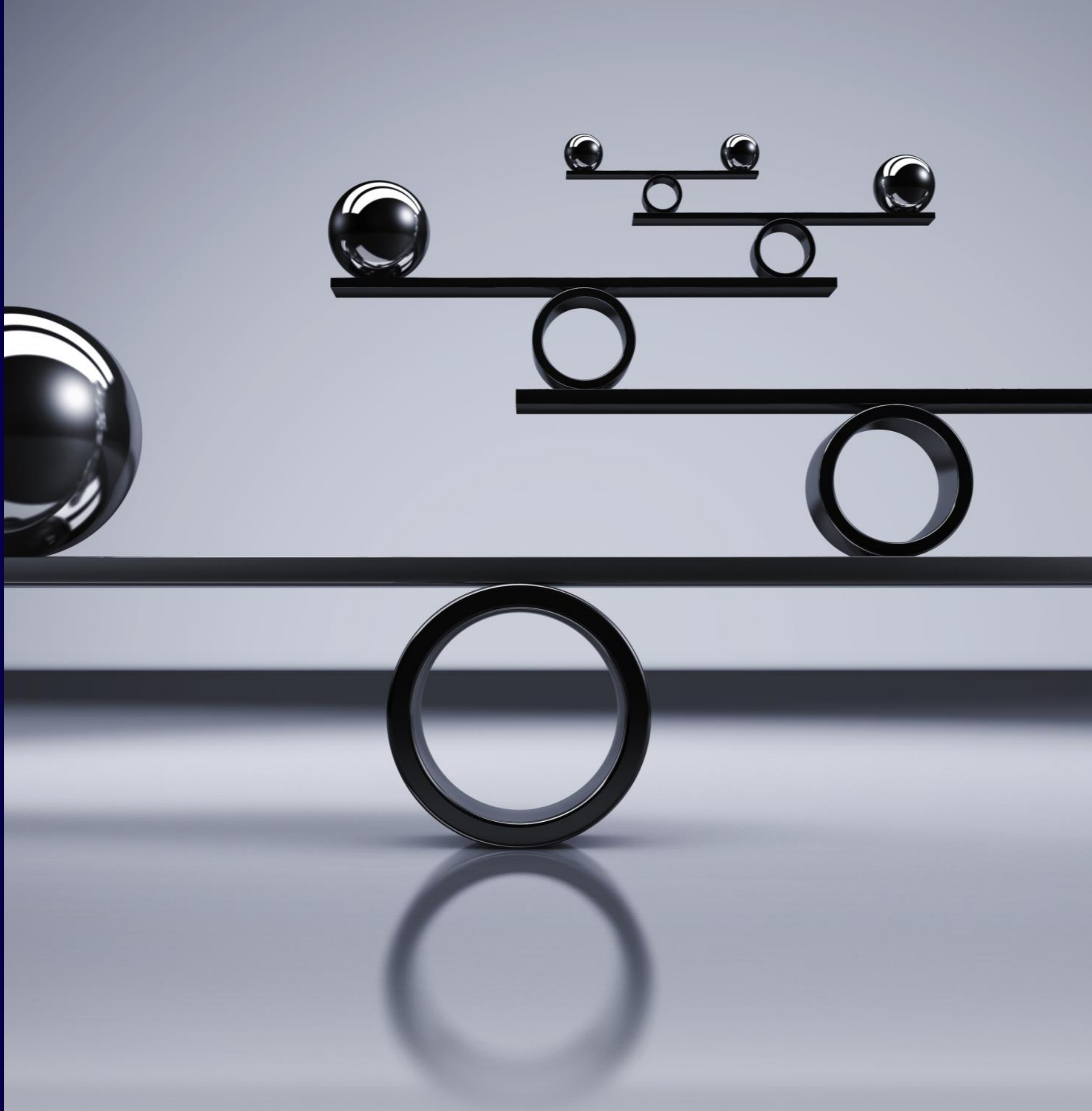
Optimization Challenges

- We now need to educate engineering about costs.
- A lot of the challenge comes from Kubernetes and how containers are deployed.
- Collaboration issues can hamper optimization
- Lack of Corporate direction(buy in and policy)



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Container Spending To Do List

- Setup cost allocation practices and policy examples to govern container spending
- Setup QOS
- Use Best Effort allocation
- Use Burstable allocation
- Understand Static vs Runtime costs
- Labeling and Namespace
- Rightsizing
- Discounts

Container Services

- Amazon Elastic Kubernetes Service (Amazon EKS)
- Google Kubernetes Engine (GKE)
- Azure Kubernetes Service (AKS)
- IBM Cloud Kubernetes Service (CKS)

Container Spending Example

- Billing Hierarchy: Organizations, folders, projects, normalizing them with cross-cloud concepts: Linked Accounts, Tags, Subscriptions, etc.
- Resources: Compute cores, RAM, GPU, TPU, Load Balancers, Persistent Disk, Custom Machines, Network Egress
- Namespaces: labeling specific, isolated containers
- Labels: Teams, cost centers, app names, environment, and more

https://www.finops.org/projects/calculating-container-costs/

Test Tips

- Container costs are difficult to manage
- Traditional FinOps cost allocation doesn’t work.
- Multiple Containers/Apps
- Use Container Classes
- Labeling and Namespace

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FinOps Terminology

Understanding common terms with FinOps

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Terminology Breakdown for the Exam

- Cloud Cost Management Terminology
- Business Terminology
- Public Cloud Terminology
- Software Development & Operations Terminology
- Finance & Accounting Terminology

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Terminology

- Cost Allocation
- Cloud Waste (Wasted)
- Right Sizing
- On Demand Rate
- Spot Rate
- Rightsizing
- Rate Reduction

https://www.finops.org/resources/terminology/

Terminology

- Cost Avoidance
- Cost Savings
- Chargeback
- Savings Potential
- Savings Realized
- Reservations
- Commitments
- Reservation Waste

https://www.finops.org/resources/terminology/

Terminology

- Covered Resources
- Blended Rates
- Unblended Rates
- Amortized Costs
- Fully Loaded Costs
- Matching Principles
- Cost of Capital/WAAC
- CAPEX and OPEX

https://www.finops.org/resources/terminology/

Test Tips

- Terms are important on the exam.
- Review the terminology list on the FinOps Foundation

Cloud Language and Business Language

Understanding how Cloud Provider Naming correlate to business language

Providers and Service Names

- AWS
- Azure
- GCP

Comparing Service Provider Terms/Language

VM Comsumption	Provider	Provider Name for Reserved Instance
Virtual Machine Reservation	AWS	Reservations/Savings Plans
Virtual Machine Reservation	Azure	Reservations
Virtual Machine Reservation	GCP	Committed Use Discounts/Flat Rate Plan

Comparing Service Provider Terms/Language

Compare AWS and Azure services to Google Cloud

Send feedback

Last updated: August 31, 2021

This table lists generally available Google Cloud services and maps them to similar offerings in Amazon Web Services (AWS) and Microsoft Azure. You can filter the table with keywords, such as a service type, capability, or product name. We welcome your feedback to help us keep this information up to date!

Filter by entering a product name or characteristic.

Service category	Service type	Google Cloud product	Google Cloud product description	AWS offering	Azure offering
App Modernization	CI/CD	Cloud Build	Build, test, and deploy on Google Cloud serverless CI/CD platform	AWS CodeBuild, AWS CodeDeploy, AWS CodePipeline	Azure DevOps, Github Enterprise
App Modernization	Multi-cloud	Anthos	Anthos is a managed application platform that extends Google Cloud services and engineering practices to your environments so you can modernize applications faster and establish	AWS Outposts	App Modernization with Microsoft Azure, Azure Arc

https://cloud.google.com/free/docs/aws-azure-gcp-service-comparison

Test Tips

- Use a common language across the business between business units. (lexicon)
- Define Terms clearly.

Section Summary

Section 2 : FinOps Fundamentals

2

Section

Section Summary

- FinOps is a process and a set of best practices to bring the following aspects to companies and stakeholders:
- FinOps is meant to help your enterprise to prioritize ongoing optimization of your cloud costs and reduce waste.
- Provide a better ROI. (Profit)
- The practice of bringing financial accountability to the variable spend model of cloud, enabling teams to make business trade-offs between spend, cost and quality.
- The shift from CAPEX to OPEX financing and cash flow models using cloud
- Cloud TCO (total cost of ownership) is the total costs associated with your new cloud technology which is often compared to the total costs of your former server or data center deployment.
- An ROI calculation quantifies both the costs and the expected benefits of a specific project over a specific timeframe. (3-5 years)
- Terminology to know. (Business, Tech, etc)
- Standardize Business Language

TECH COMMANDERS
ADVANCING YOUR IT SKILLS

Section Review Questions

Section 2 : FinOps Fundamentals

TECH COMMMANDERS
ADVANCING YOUR IT SKILLS

Review Questions

Which of following answers would be true around Devops and Cloud resources? (Select Three)

- DevOps consumes resources faster and unpredictably
- Cost is not always a concern with DevOps engineers. Cost vs Performance
- User deploy Hard Coded Resources (IaC)
- Traditional FinOps works all the time.
- DevOps should not be involved with FinOps

Review Questions

Which of following answers would be true around Devops and Cloud resources? (Select Three)

- DevOps consumes resources faster and unpredictably
- Cost is not always a concern with DevOps engineers. Cost vs Performance
- User deploy Hard Coded Resources (IaC)
- Traditional FinOps works all the time.
- DevOps should not be involved with FinOps

Review Questions

The FinOps Lifecycle has three phases. What are they? (Select Three)

- Inform
- Exploit
- Optimize
- Evolve
- Operate

Review Questions

The FinOps Lifecycle has three phases. What are they? (Select Three)

- Inform
- Exploit
- Optimize
- Evolve
- Operate

Review Questions

Which of the following answers would be correct around Cloud Computing and how resources are consumed? (Select Two)

- Cloud computing introduces new ways to consume and pay for IT services
- The shift from CAPEX to OPEX financing and cash flow models using cloud
- The shift from OPEX to CAPEX financing and cash flow models using cloud
- Cloud computing introduces new culture and new cash flow methods

Review Questions

Which of the following answers would be correct around Cloud Computing and how resources are consumed? (Select Two)

- Cloud computing introduces new ways to consume and pay for IT services
- The shift from CAPEX to OPEX financing and cash flow models using cloud
- The shift from OPEX to CAPEX financing and cash flow models using cloud
- Cloud computing introduces new culture and new cash flow methods

Section 3 – FinOps Teams and Members

Understanding the teams and structure

Section Overview

WHAT IS A FINOPS TEAM

FINOPS TEAM ROLES

CENTRALIZED TEAMS OR DECENTRALIZED?

ORGANIZATION CHANGES AND ADOPTION

CULTURE OF FINOPS

MOTIVATIONS

SECTION SUMMARY

SECTION REVIEW QUESTIONS

TECH COMMANDERS
ADVANCING YOUR IT SKILLS

What is a FinOps Team

Understanding Teams

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- FinOps effectively brings Finance, Tech and Business teams together to improve business efficiency while bringing value by using cloud technologies.
- The art of FinOps is to bring this combination together with a change in team culture and internal processes.
- Collaborative and Centralized

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The FinOps team should be responsible for:

- Defining enterprise cloud usage strategy
- Defining and adjusting enterprise cloud budgets
- Setting up the cloud usage practices and guidelines
- Reviewing the data and determining corrective actions as needed

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- Cloud costs need to be visible to everyone.
- Finance should not be the only BU managing costs
- In the cloud we use a shared responsibility model.

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FinOps Team Structure

Finance/Procurement

Executives

Stakeholders

Engineering/Prod/Dev

Rate Negotiations

Configuration Mgmt.

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FinOps Team structure should be considered for the enterprise

- Stakeholders
- Centralization
- Meet Business Goals
- Roles and Responsibility

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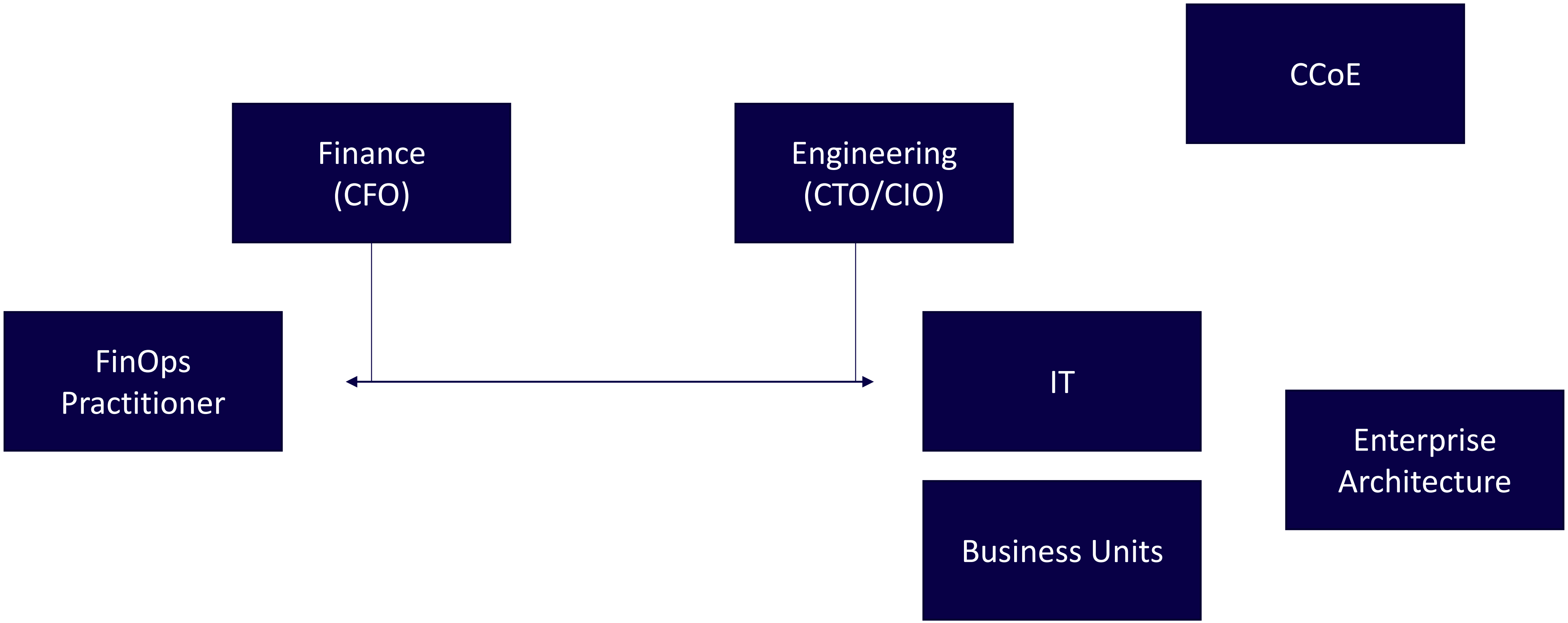

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Establishing a team by identifying specific members from:

- Cloud Center of Excellence (CCoE)
- Sponsor(s)
- Engineering
- Finance
- Application Owners
- DevOps
- And more business units as needed

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FinOps Team Should be placed

- FinOps Team should be under the CFO/COO in most enterprises.
- Should be aligned with engineering/development
- Should be collaborative with Finance and Engineering to meet business goals.

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Evaluating Reserved Instances

- A Net Present Value (NPV) analysis could be used for evaluating Reserved Instances

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Test Tips

- FinOps brings together Business, Finance and Tech.
- FinOps teams and members need to be defined.
- Roles should be defined for members.
- FinOps should be under the CFO/Finance

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FinOps Team Roles

Understanding Team Roles

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Establishing a FinOps Team. Possible members should be.

→ CTO

→ CIO

→ CFO

→ Directors of Engineering /Ops/Dev

→ Financial Controllers

→ Engineering Managers

→ Project Managers

→ Enterprise Architects

→ Cloud Engineer/Architects

Personas

- Roles are Personas
- Everyone who is part of a FinOps organization should have primary goals
- FinOps Practitioner - Primary Goal is to drive best practices into the organization through education, standardization, and cheerleading

Persona Factors

FinOps Practitioner

- Objectives –Drive Change
- Frustrations – Lack of Buy in
- Key Metrics - Forecasts
- FinOps Benefits –Aligns accountability to users.

Financial analyst and controller - translates expectations and goals from the finance team. It can be CFO or somebody from the CFO office.

Cloud Practitioner – sets cloud usage best practices, educates engineers how to properly use clouds and explains why cloud expenses should be another KPI in R&D.

Executives - Senior VP or a C-level person who controls how the collaboration works and reviews results.

Engineers – uses clouds according to the best practices and budgets, report if there are any escalations.

Test Tips

- Each role in a FinOps based organization should be specified.
- Part of the FinOps team
- Roles = Personas
- FinOps Practitioner - Drive best practices into the organization through education, standardization, and cheerleading

Test Tips

- Encourage teams to act by using chargeback/show back for cloud resources.

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Centralized or Decentralized

Should a Team be centralized or Not?

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A FinOps Team should be centralized. Why?

- Increases collaboration
- Increases culture
- Removes duplication of efforts
- Streamlines planning and decisions
- Removes bias and reduces disagreements
- Alignment with business goals

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Centralized Team

Organization Changes and Adoption

Change for successful adoption of FinOps

Organization Changes and Adoption

Change needs to occur to an organization for FinOps to become successful.

Some of the changes for adoption would be

1. Executive Buy In
2. Obtaining Funding to establish a FinOps practice. (People, Training and Tools)
3. Assigning Responsible Members (Key Personas) to drive adoption

Test Tips

- For change to happen it must occur with proper organization change and direction.

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Culture of FinOps

Culture of FinOps

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Establishing a FinOps Culture

An organization's FinOps culture embraces a long-term roadmap of transformation and continuous improvement across all the FinOps Domains at each stage of FinOps Maturity. (FinOps Foundation)

- Cultural Practice (proactive)
- Taking Ownership
- Collaborative
- Best Practices

Motivations

Culture of FinOps

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FinOps practitioners must build reporting and processes that reduce the need for both finance and engineering teams (efficiency)

Common Language enables teams to familiarize themselves with a common set of terms that we understand. (remove barriers)

Every FinOps stake holder will specific motivations. Product Manager, Executives, Engineers.

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Motivations

FinOps teams will promote both ‘cost avoidance,’ which relates to usage, and ‘cost optimization.

Most of the actions required for cost avoidance are engineer-dependent.

Each stakeholder has different motivations.

For example, an engineer will not be concerned about cost whereas an executive will care more amortizing costs to teams.

Test Tips

- Executives will be motivated by amortizing costs.
- Product Managers will want to ensure support models are operational.

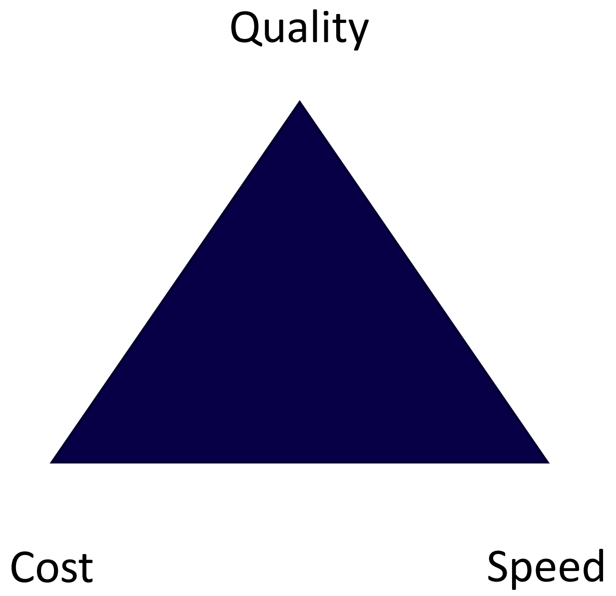
FinOps Triangle

Trade - Offs

The FinOps Iron Triangle is like the project triangle

Trade Offs (Compromise) – Quality, Speed and Cost

- Speed: How fast we would like to deliver.
- Quality: Number of bugs, performance, ease of use, etc…
- Cost: Total cost for development and maintenance



Section Summary

Section 3 : FinOps Teams and Motivations

TECH COMMMANDERS
ADVANCING YOUR IT SKILLS

3

Section

Section Summary

- FinOps effectively brings Finance, Tech and Business teams together to improve business efficiency while bringing value by using cloud technologies.
- A FinOps Team should be centralized for numerous reasons such as collaboration, removes redundancy, standards, etc.
- FinOps Practitioner - Primary Goal is to drive best practices into the organization through education, standardization, and cheerleading
- The FinOps team should be responsible for defining enterprise cloud usage strategy, defining and adjusting enterprise cloud budgets and much more.
- Some of the changes for adoption would be Executive Buy In, Obtaining Funding to establish a FinOps practice and Assigning Responsible Members (Key Personas) to drive adoption
- An organization’s FinOps culture embraces a long-term roadmap of transformation and continuous improvement across all the FinOps Domains at each stage of FinOps Maturity.

TECH COMMANDERS
ADVANCING YOUR IT SKILLS

Section Review Questions

Section 3 : FinOps Teams and Motivations

TECH COMMMANDERS
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Review Questions

The FinOps Triangle has three constraints. What are they? (Select One)

- Quality, Speed and Cost
- Quality, Budgets and Expenses
- Speed, Costs, Budget
- Speed, Costs, Availability

Review Questions

The FinOps Triangle has three constraints. What are they? (Select One)

- Quality, Speed and Cost
- Quality, Budgets and Expenses
- Speed, Costs, Budget
- Speed, Costs, Availability

Review Questions

Changes needs to occur to an organization for FinOps to become successful. Which of the following would NOT be a change needed? (Select One)

- Obtain Executive Buy In
- Obtaining Funding to establish a FinOps practice.
- Assigning Responsible Members to drive adoption.
- Assign a Project Management Team to handle the FinOps transition.

Review Questions

Changes needs to occur to an organization for FinOps to become successful. Which of the following would NOT be a change needed? (Select One)

- Obtain Executive Buy In
- Obtaining Funding to establish a FinOps practice.
- Assigning Responsible Members to drive adoption.
- Assign a Project Management Team to handle the FinOps transition.

Review Questions

Everyone that is part of the FinOps Organization should have a primary goal.

True or False? (Select One)

- True
- False

Review Questions

Everyone that is part of the FinOps Organization should have a primary goal.

True or False? (Select One)

- True
- False

Review Questions

Which of the following reasons would NOT be true regarding why a FinOps team should be selected? (Select One)

- Increases collaboration
- Removes culture
- Removes duplication of efforts
- Streamlines planning and decisions

Review Questions

Which of the following reasons would NOT be true regarding why a FinOps team should be selected? (Select One)

- Increases collaboration
- Removes culture
- Removes duplication of efforts
- Streamlines planning and decisions

Section 4 – FinOps Capabilities

Understanding functional areas of your FinOps practice

Section Overview

FINOPS

DOMAINS AND CAPABILITIES

Six Pillars of FinOps

Best Practices

Aligning Teams to Resources

Section Summary

Section Review Questions

FinOps Domains and Capabilities

What are FinOps Domains and Capabilities?

FinOps Domains represents a sphere of activity or knowledge.

Each FinOps Domain consists of FinOps Capabilities.

Capabilities are "Functional Activities" to meet the demands of a FinOps practice/domain.

Domains and Capabilities

https://www.finops.org/framework/domains/

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Domain -
Understanding Cloud Usage and Cost

Capability

Cost Allocation (Metadata/Hierarchy)
Forecasting

Capability

Forecasting
Resource Utilization

FinOps Foundation Domains

https://www.finops.org/img//resources/FinOps%20Poster_011021.pdf

Domains

Understanding Cloud Usage & Cost

- Cost allocation (metadata & hierarchy)
- Manage Anomalies
- Manage Shared costs
- Data analysis & showback
- Data ingestion & normalization
- IT asset management

Performance Tracking & Benchmarking

- Measure efficiency
- Trend & variance analysis
- Forecasting
- Budget management
- Budget thresholds & alerts
- Measure unit costs

Real-Time Decision Making

- Persona identification
- Identify anomalies
- Budget alerts
- Identify optimization opportunities
- Timely analysis curated for stakeholders
- Business case velocity

Cloud Rate Optimization

- Manage commitment based discounts
- Volume & usage based discounts
- Pricing model analysis
- License management
- Leverage unused cloud provider capacity

Cloud Usage Optimization

- Workload management & automation
- Resource Utilization & Rightsizing
- Cost avoidance
- Adopt modern cloud provider technologies
- Architecture modernization
- Application decomposition
- Observability

Organizational Alignment

- Establish FinOps culture
- Chargeback & IT finance integration
- Policy & governance
- Continual learning, knowledge sharing & advocacy programs
- Socialize innovation for culture change & adoption
- Modernize processes to align with cloud cost management

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FinOps Capabilities

- Cost Allocation (Metadata & Hierarchy)
- Data Analysis and Showback
- Managing Shared Cost
- Data Ingestion & Normalization
- Managing Anomalies
- Forecasting
- Measuring Unit Costs

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FinOps Capabilities

- Measuring Unit Costs
- Managing Commitment Based Discounts
- Resource Utilization & Efficiency
- Forecasting
- Budget Management
- Managing Anomalies

Domain - Performance Tracking & Benchmarking

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FinOps Capabilities

- Cost Allocation (Metadata & Hierarchy)
- Data Analysis and Showback
- Managing Anomalies
- Managing Commitment Based Discounts
- Resource Utilization & Efficiency
- Measuring Unit Costs
- Establishing a FinOps Decision & Accountability Structure

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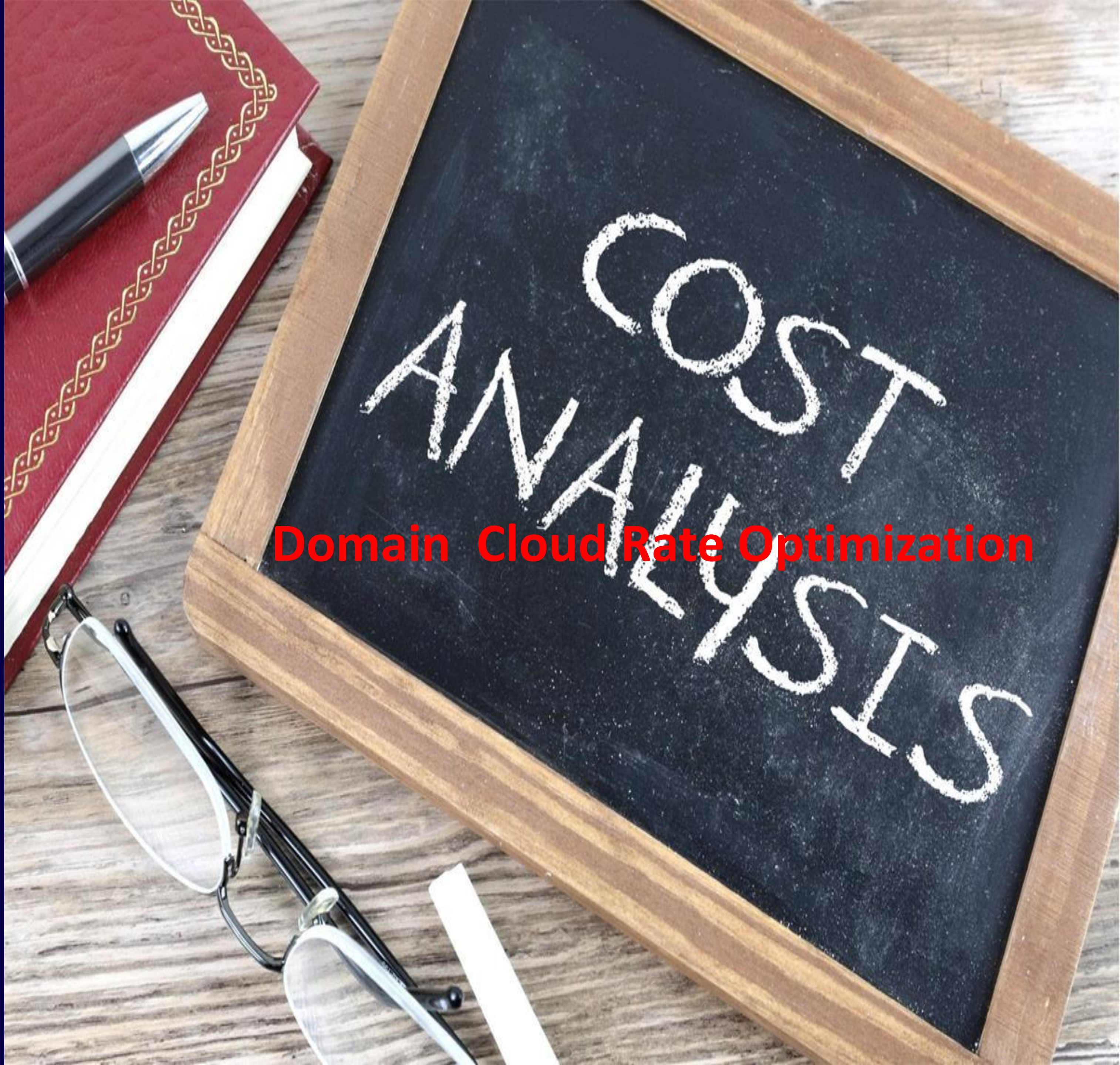
FinOps Capabilities

- Data Analysis and Showback
- Managing Commitment Based Discounts



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FinOps Capabilities

- Data Analysis and Showback
- Onboarding Workloads
- Resource Utilization & Efficiency
- Workload Management & Automation

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FinOps Capabilities

- Establishing FinOps Culture
- Managing Shared Cost
- Chargeback & IT Finance Integration
- Data Analysis and Showback
- Budget Management
- FinOps Education & Enablement
- Establishing a FinOps Decision & Accountability Structure
- Cloud Policy & Governance
- IT Asset Management Integration

Domain Organizational Alignment

Test Tips

- FinOps Domains represents a specific sphere of activity or knowledge.
- Each FinOps Domain consists of FinOps Capabilities and Capabilities can overlap between Domains.
- Forecasting for example can be accomplished under both the
- Understanding Cloud Usage and Cost & Performance Tracking & Benchmarking Domains

Six Pillars(Principles) of FinOps

What are the Six Principles and how they tie into capabilities?

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Teams need to collaborate.

Decisions are driven by the business value of cloud.

Everyone takes ownership of their cloud usage.

FinOps reports should be accessible and timely.

A centralized team drives FinOps.

Take advantage of the variable cost model of the cloud.

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Teams need to collaborate.

- Collaboration and Communication
- Define Governance and Control
- Increase Cost Efficiency and Performance

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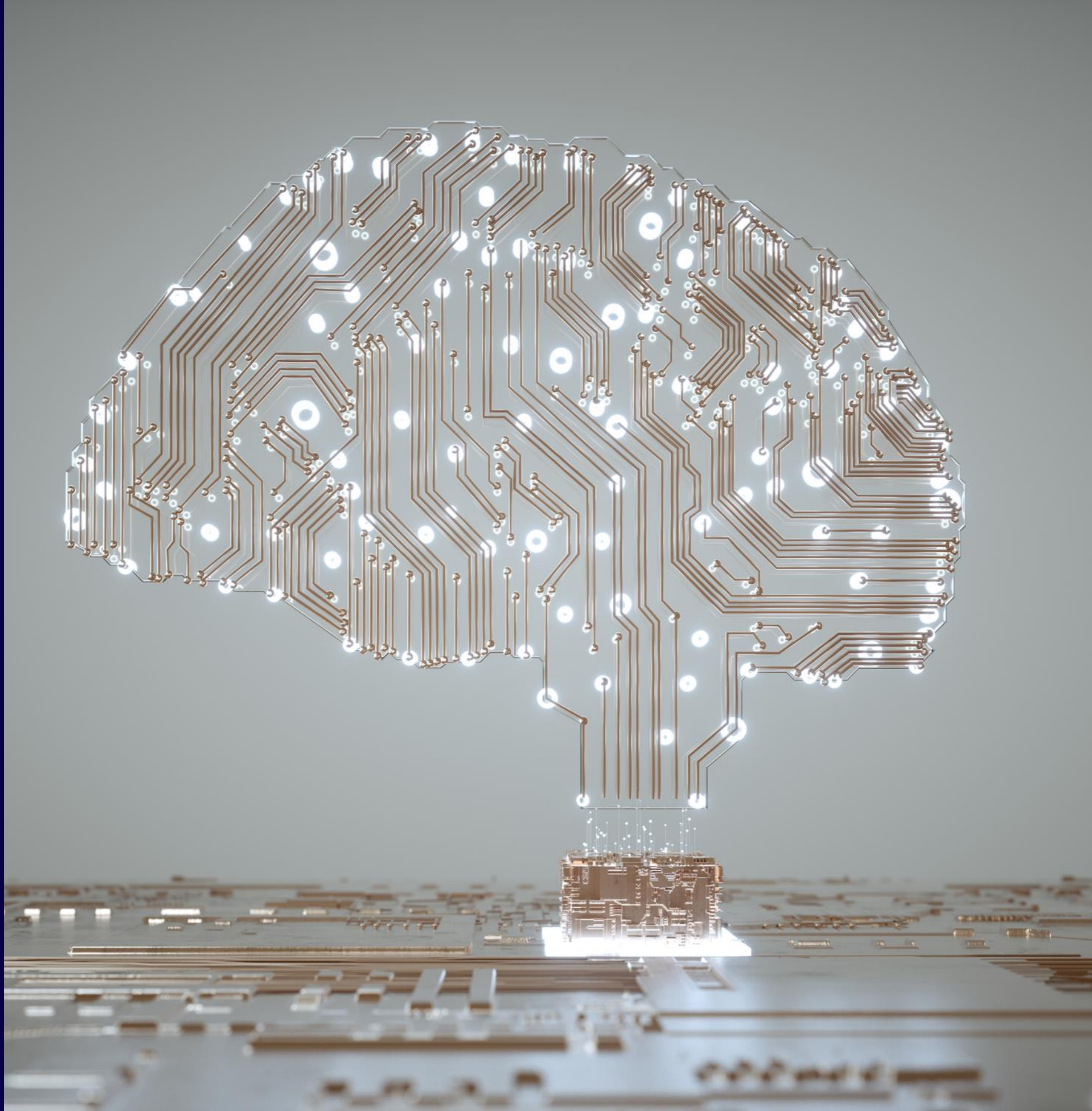
Decisions Are Driven by the Business Value of Cloud

- Maximize the value created by the spend.
- Business Metrics



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Everyone Takes Ownership of Their Cloud Usage

- Responsibility
- Chargeback
- Edge

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FinOps Reports Should Be Accessible and Timely

- Real Time Decisions
- Alerts and Monitoring
- Cost Avoidance
- True Costs Focus on Clean Decisions.
- FinOps decisions are based on fully loaded and properly allocated costs

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A Centralized Team Drives FinOps

- Drive value thru education, standardization, and cheerleading
- Cost Centralization
- Benchmarking

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Take Advantage of the Variable Cost Model of the Cloud

- Foreword looking (forecasting)
- Reserved Instances
- Rightsizing
- Volume Discounts

Test Tips

- Decisions are made with data
- Ownership of cloud costs should go to the edge
- FinOps decisions are based on fully loaded and properly allocated costs

Best Practices

Best Practices

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Inform - Allocate, benchmark, and forecast cloud service needs.

Optimize –Compare the needs of the company with discounts for time and reserved usage in order to get the better rates.

Operate - Continuously evaluate the performance of cloud services against business objectives for speed, quality and cost.

Best Practices

- Cloud stakeholders need to agree on a comprehensive organizational grouping and tagging regime.
- Cloud as a business value generator
- Build Communication channels for collaboration (remove silos)
- Develop a common vocabulary
- Centralize a FinOps organization.
- Setup Kubernetes shared cost allocation models.

Best Practices

- Forecasts for a 3–12-month timeframe.
- Develop forecasts with the help of machine learning tools.
- Develop scorecards aimed at comparing teams.
- Use Real Time Reports
- Expand Reporting and Forecasting
- Shift Left Cost Accountability
- Understand Cost Influence - Cost avoidance and Rate reduction.

Aligning Teams to Resources

Ensuring the right people manage the right resources

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Managing People and Resources

- Resources – VMs, Services, Storage, etc
- Teams – Business Units, Define stakeholders, structure, train

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Teams (people)

- Executive buy-in for FinOps and its practices and processes is required.
- Teams need to collaborate
- Continuously improve for efficiency and innovation.
- Everyone takes ownership of their cloud usage.
- Shift Left
- Technical teams balance performance and costs
- Engineers and operations teams stay focused on usage optimization.

Resources

- Enhance visibility into resource usage
- Instance types – Reserved, On Demand, Spot, etc.
- Volume/Custom Discounts with cloud providers

Test Tips

- Realize the right team collaboration can provide for results.
- Engineers and operations teams stay focused on usage optimization.
- Executive buy-in for FinOps and its practices and processes is required.

Section Summary

Section 4 : FinOps Capabilities

TECH COMMMANDERS
ADVANCING YOUR IT SKILLS

4

Section

Section Summary

- Each FinOps Domain consists of FinOps Capabilities.
- Capabilities are “Functional Activities” to meet the demands of a FinOps practice/domain.
- Executive buy-in for FinOps and its practices and processes is required.
- Managing Anomalies is under the Understanding Cloud Usage and Cost
- A Centralized Team Drives FinOps drives value thru education, standardization, and cheerleading as well as Cost Centralization and Benchmarking
- Inform - Allocate, benchmark, and forecast cloud service needs.
- Optimize –Compare the needs of the company with discounts for time and reserved usage in order to get the better rates.
- Operate - Continuously evaluate the performance of cloud services against business objectives for speed, quality and cost.

Section Review Questions

Section 4 : FinOps Capabiities

TECH COMMANDERS
ADVANCING YOUR IT SKILLS

Review Questions

Each FinOps Domain consists of one or more FinOps Capabilities. True or False (Select One)

- True
- False

Review Questions

Each FinOps Domain consists of one or more FinOps Capabilities. True or False (Select One)

- True
- False

Review Questions

Which of the following would be the main goals for the Inform Phase? (Select One)

- Allocate, benchmark, and forecast cloud service needs.
- Compare the needs of the company with discounts for time and reserved usage in order to get the better rates.
- Continuously evaluate the performance of cloud services against business objectives for speed, quality and cost.

Review Questions

Which of the following would be the main goals for the Inform Phase? (Select One)

- Allocate, benchmark, and forecast cloud service needs.
- Compare the needs of the company with discounts for time and reserved usage in order to get the better rates.
- Continuously evaluate the performance of cloud services against business objectives for speed, quality and cost.

Review Questions

Which of the following would be true regarding the principle “FinOps Reports Should Be Accessible and Timely? (Select Three)

- Real Time Decisions are not needed
- Alerts and Monitoring should be setup
- Cost Avoidance is never an objective
- True Costs Focus on Clean Decisions.
- FinOps decisions are based on fully loaded and properly allocated costs

Review Questions

Which of the following would be true regarding the principle “FinOps Reports Should Be Accessible and Timely? (Select Three)

- Real Time Decisions are not needed
- Alerts and Monitoring should be setup
- Cost Avoidance is never an objective
- True Costs Focus on Clean Decisions.
- FinOps decisions are based on fully loaded and properly allocated costs

Section 5 – FinOps Lifecycle

Inform, Optimize and Operate

Section Overview

THE FINOPS LIFECYLE

LIFECYCLE PHASE - INFORM

LIFECYCLE PHASE - OPTIMIZE

LIFECYCLE PHASE - OPERATE

WHITEBOARD - CONSIDERATIONS

SECTION SUMMARY

SECTION REVIEW QUESTIONS

TECH COMMANDERS
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FinOps Lifecycle Overview

Lifecycle and Phase Overview

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- The FinOps lifecycle follows the three phases of the Crawl, Walk, Run
- Inform
- Optimize
- Operate

INFORM

- Understand your organizations resource consumption thru context, data, budgets and Forecasting.

OPTIMIZE

- Setting your goals and targets by determining you cloud costs thru strategy, tracking, cost breakdown and cost growth boundary.

OPERATE

- Aligning organization teams to business goals thru drive and action. Focus is on getting results for the organization.

FinOps Lifecycle

- The FinOps journey consists of three iterative phases — Inform, Optimize and Operate.
- FinOps is a Continuous Journey
- Actions and Goals?

- The official FinOps Lifecycle Phases documentation is listed here.

https://www.finops.org/framework/phases/

Please review before taking the exam

FinOps Lifecycle - Inform

Inform Phase

INFORM

- Understand your organizations resource consumption thru context, data, budgets and Forecasting.

OPTIMIZE

- Setting your goals and targets by determining you cloud costs thru strategy, tracking, cost breakdown and cost growth boundary.

OPERATE

- Aligning organization teams to business goals thru drive and action. Focus is on getting results for the organization.

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- Forecasting
- Visibility
- Allocation
- Benchmarking
- Budgeting

Inform Phase - What are the primary Goals?

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FinOps Lifecycle – Optimize Phase

Optimize Phase

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INFORM
- Understand your organizations resource consumption thru context, data, budgets and Forecasting.

OPTIMIZE
- Setting your goals and targets by determining you cloud costs thru strategy, tracking, cost breakdown and cost growth boundary.

OPERATE
- Aligning organization teams to business goals thru drive and action. Focus is on getting results for the organization.

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- Defining clear service level objectives (SLOs)
- Improve Reserved Instance Usage
- Right Sizing
- Eliminate Waste
- Automate Scaling

Optimize Phase - What are the primary Actions?

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FinOps Lifecycle – Operate Phase

Operate Phase

INFORM

- Understand your organizations resource consumption thru context, data, budgets and Forecasting.

OPTIMIZE

- Setting your goals and targets by determining you cloud costs thru strategy, tracking, cost breakdown and cost growth boundary.

OPERATE

- Aligning organization teams to business goals thru drive and action. Focus is on getting results for the organization.

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- Define Customized Processes (Onboarding, Deployments, etc)
- Automate Everything
- Evaluate Team Performance
- Establish new Goals that improve speed, quality and performance
- Automate Scaling

Operate Phase - What are the primary Actions?

Test Tips

- Onboarding should be clearly defined in the Operate Phase
- Processes should be documented
- Automation should be planned and implemented for processes
- New or Existing Cloud Applications/Services.

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Section Summary

Section 5 : FinOps Lifecycles

TECH COMMMANDERS
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5

Section

Section Summary

- The FinOps journey consists of three iterative phases — Inform, Optimize and Operate.
- The FinOps lifecycle follows the three phases of the Crawl, Walk, Run
- Inform – Setting Goals - Understand your organizations resource consumption thru context, data, budgets and Forecasting.
- Optimize – Achieving your goals and targets by determining you cloud costs thru strategy, tracking, cost breakdown and cost growth boundary.
- Operate –Aligning organization teams to business goals thru drive and action. Focus is on getting results for the organization.

TECH COMMMANDERS
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Section Review Questions

Section 5 : FinOps Lifecycles

TECH COMMANDERS
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Review Questions

- Onboarding should be clearly defined in the Operate Phase? True or False (Select One)

  - True
  - False

Review Questions

- Onboarding should be clearly defined in the Operate Phase? True or False (Select One)

  - True
  - False

Review Questions

- Which of the following are primary goals? (Select Three)

  - Visibility
  - Authorization
  - Benchmarking
  - Performance
  - Forecasting

Review Questions

- Which of the following are primary goals? (Select Three)

- Visibility
- Authorization
- Benchmarking
- Performance
- Forecasting

Section 6 – Cloud Resource Pricing and Billing

Understanding Provider Requirements and Processes

Section Overview

- COMPUTE ECONOMICS AND PRICING
- CAPACITY RESERVATIONS AND RESERVED INSTANCES
- CLOUD BILLING
- CLOUD SPEND MANAGEMENT AWS
- AWS PRICING MODELS
- DEMO - AWS COST EXPLORER
- DEMO - AWS SPOT INSTANCES
- AZURE PRICING MODELS
- DEMO - AZURE VMS
- GCP PRICING MODELS
- DEMO - GCP INSTANCE DEMO
- SECTION SUMMARY
- SECTION REVIEW QUESTIONS

Cloud Economics and Pricing

Understanding the pricing models in Cloud

Economics of Cloud

Cloud economics is the study of cloud computing's costs and benefits and the economic principles that underpin them. As a discipline, it explores key questions for businesses:

- What is the return on investment (ROI) of migrating to the cloud or switching current cloud providers?
- And what is the total cost of ownership (TCO) of a cloud solution versus a traditional on-premises solution?

Economics of Cloud

Pricing in the cloud is based on various factors.

- Provider
- Services (Compute, Data, Network, Storage)
- Bandwidth used
- Connection used
- Support

Economics of Cloud

Characteristics of Cloud Computing

- Broad network access.
- On-demand self-service.
- Resource pooling.
- Measured service.
- Rapid elasticity.

Capacity Reservations and Reserved Instances

Understanding the pricing models in Compute

Reduce Cloud Spending

- Reserved Instances provide up to a 75% discount
- Reserved Instances provides a “predictable” cost model
- With RIs, you commit in advance for usage which in return means a lower price
- With reservations, you can choose to pay with no-upfront, partial upfront or all upfront
- More upfront payment, the bigger discount

EC2 Pricing Cost Models

There are four pricing models for Amazon EC2 instances:

- On-Demand Instances – Provides for instant use with no commitment
- Reserved Instances – Provides for cost savings with commitment
- Spot Instances – Available AWS capacity at a significant discount for short term
- Dedicated Hosts – Use for license bound/compliance applications

Reduce Cloud Spending

Compute Savings Plans for Amazon EC2 | Compute Savings Plans for AWS Fargate | Compute Savings Plans for AWS Lambda | EC2 Instance Savings Plans

Compute Savings Plans for Amazon EC2

Compute Savings Plans apply to EC2 Instance usage regardless of instance family, size, AZ, AWS Region, OS or tenancy.

Select terms for your Compute Savings Plans

Term length: 1 year

Payment options: No Upfront

https://aws.amazon.com/savingsplans/pricing/

Select a region, operating system, and tenancy to view rates

Region: US East (Ohio)

Operating system: Linux

Tenancy: Shared

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Cloud Billing

Understanding Billing in Cloud

Billing

- Cloud Billing is complex and can vary widely between providers.
- Cloud Billing is the process of generating bills from the resource usage data which is done via predefined billing policies.
- Providers use what is effectively a SKU for each service. For example, every Vm instance template and individual region/zone you may deploy in have its own billing characteristics.
- AWS has over 200 Individual SKUS
- It is very important to choose the right billing model (Reserved, Spot, Savings)

Billing

- The rate for a specific resource type in a specific row may be different than the same resource type in a different resource time period.
- Paying less for what you use is called rate reduction
- To save money you can Shutdown or stop resources over nights or weekends
- Cloud billing is the process of generating bills from the resource usage data using a set of predefined billing policies.

Billing

- A central team is better equipped to manage RI and CUD commitments due to their ability to look across the entire cloud estate and the complexity of managing a large portfolio of commitments.
- Rightsizing VMS and scale down the number of resources during off peak consumption.
- It’s a best practice to terminate resources that are not being used.
- A FinOps professional needs to be able to view and understand raw data

Reducing Your Cloud Bill

- Use the right provider tools and or third-party tools
- Aggregate the usage data and to identify waste
- Identify Rightsizing opportunities and correct instance types
- Pay upfront when appropriate. (Usage Discounts)
- Terminate resources not used with automation.

Cost Allocation Methods

Two Methods are

- Tagging and Labeling - key/value pair tags (or labels) that can be applied to a resource, as well as hierarchy-based
- Hierarchal Account Separation – Separate account that is used for only one cost center.

Tagging Methods

Three Methods are:

- Resource-level tag
- Accounts/projects/subscriptions
- Post-bill data constructs

Tagging strategies will provide structure to the resources and help you group costs.

Benchmarking Performance

Benchmarking is an important aspect in FinOps. For a practitioner to be successful we must be able to compare performance.

- A thorough understanding of an organization’s purpose and guidelines is needed.
- The ability to identify precisely dimensions, elements and processes that need benchmarking.

- Elements – Trend and Variance Analysis

Benchmarking Performance

Benchmarking you teams is a logical step to determine what teams are improving.

Test Tips

- What is the Cloud Spend Formula? Spend = Usage x Rate
- Paying less for what you use is called rate reduction
- A FinOps professional needs to be able to view and understand raw data
- Benchmarking teams to compare improvement is very important.

Test Tips

- Engineers will need to consider costs and balance performance.

Demo – Allocation Techniques

Metadata and Tagging

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Tagging

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Cloud Spend Management Tools

Managing your Cloud Spending

Reduce Cloud Spending

- Select the right instance size for your workload (Right Size)
- Use Cloudwatch, Cost Explorer to monitor costs/usage
- Use Trusted Advisor for best practices (Cost Optimization)
- Use Auto Scaling to align your resources with demand
- Consolidated Billing provides cost savings
- Shutdown or Terminate unused instances

Reduce Cloud Spending

- Reserved and Spot Instances as needed
- Consider Savings Plans
- Use Cost Allocation Tags
- Compute Optimizer
- Use Third Party Tools
- Audit your AWS deployment
- Follow Best Practices

Reduce Cloud Spending

- **Cloud Spend Formula**

  Spend = Usage x Rate

Reduce Cloud Spending

- Reserved Instances provide up to a 75% discount
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Operating system: Linux

Tenancy: Shared

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Cloud Spend Management

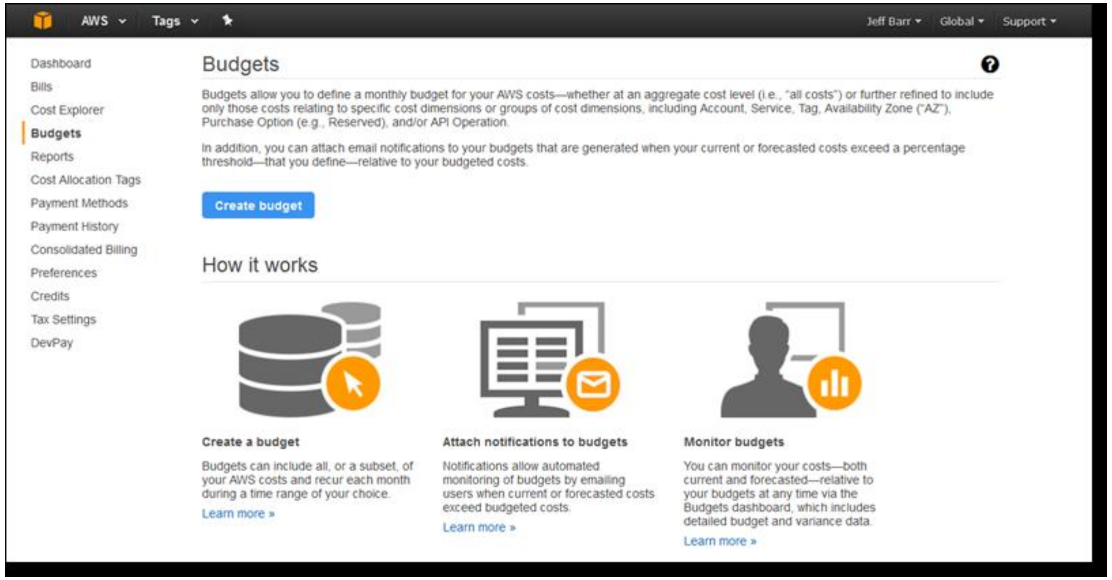
Managing your cloud spending is critical and AWS provides your enterprise tools such as:

- AWS Budgets
- Cost Explorer
- AWS Cost & Usage Reports
- Billing alerts
- Billing Preferences

Budgets

Budgets can help keeps cost in check and will send alerts

- Cost exceeds Threshold
- Forecasted to exceed
- Set Thresholds
- 2 Free budgets (Cost for exceeding)
- Up to 20,000 can be set up



Cost Explorer

Cost Explorer allows you to estimate costs and usage.

- Generate reports
- Understand cloud usage (Consumed)
- Access from Cost and Billing Management
- Access RI purchase recommendations

Budgets vs Cost Explorer

What is the real difference?

Budgets provide you the ability to set custom budgets that alert you when your costs or usage exceed (or are forecasted to exceed) your budgeted amount.

- Used to budget costs before they have been incurred

AWS Cost Explorer is an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time.

- Used to explore costs after they have been incurred

Test Tips

- Right Sizing is critical for Cost Optimization/Cloud Spend Management
- Budgets can be setup to help avoid cost overruns.
- 20,000 can be setup per account
- 2 Free Budgets, Others are at a cost

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Test Tips

- Cost Explorer is used to visualize costs after consumed.
- Budgets are used to forecasts costs and prevent overspending.

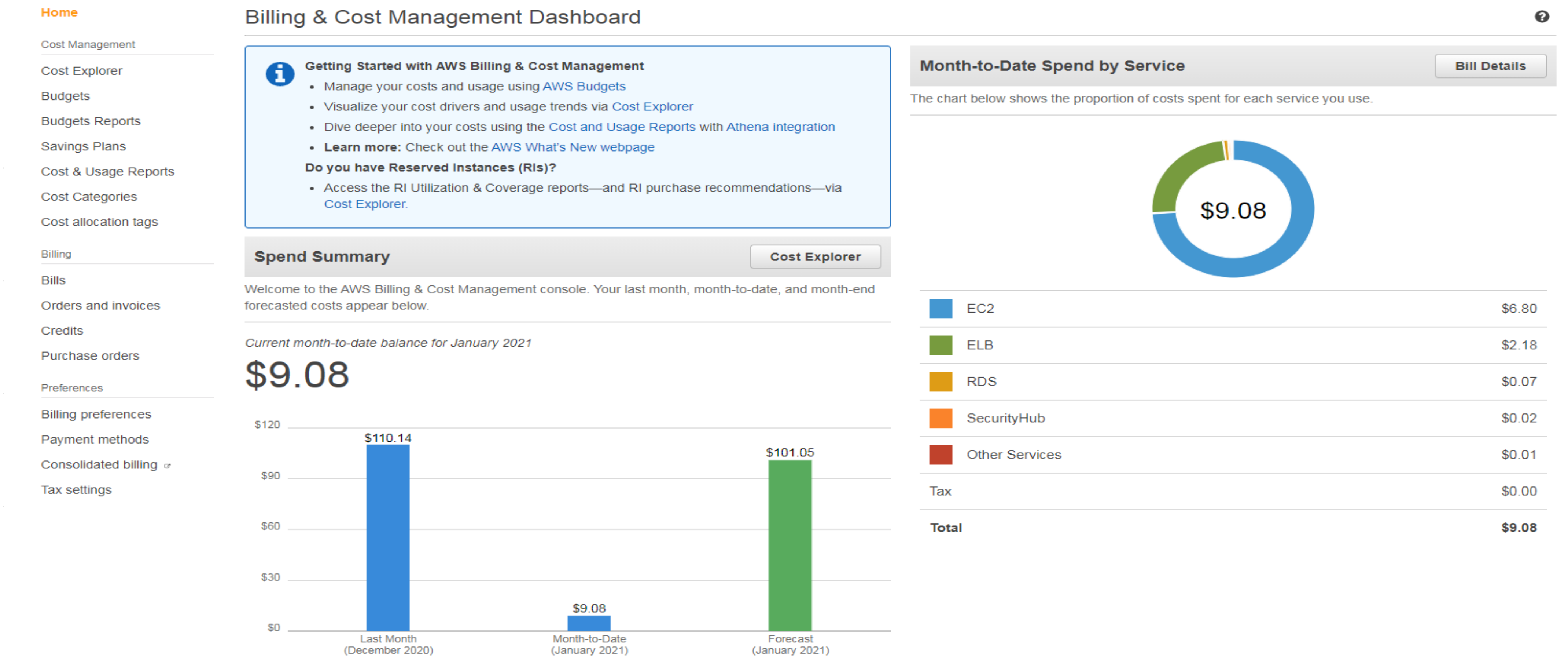
AWS Pricing Models

Understanding the pricing models in AWS

AWS Billing and Pricing Overview

- AWS provides a solid Return on Investment (ROI) when cloud spending is properly managed
- AWS provides not only flexibility and scalability in services but also pricing options.
- Pricing is based on different pricing models and you should use the proper pricing model for the proper use case.
- AWS Provides tools that provide visibility into pricing and cloud spending.

Billing and Cost Management Dashboard



Charges

- There are no charges for data transfer inbound (into AWS), data transfer outbound is charged; outbound traffic is aggregated on your bill (AWS Data Transfer Out) as /GB
- Storage is paid on a per GB basis
- Compute is paid by the minute or hour

EC2 Charging Variables

EC2 is flexible on how you deploy and how pricing is structured

Instances can be priced based on

- Region/AZ
- AMI
- Networking
- Time Deployed
- Cost Model

Amazon EC2

EC2 Charging Units

Charging Units

- The minimum unit of time that will be charged is a minute (60 seconds), but after your first minute of time, it is charged for seconds.
- Cost will vary based on region, zone and other factors

Four EC2 Pricing Options

On-Demand Instances – Provides for instant use with no commitment

Reserved Instances – Provides for cost savings with commitment

Spot Instances – Available AWS capacity at a significant discount for short term

Dedicated Hosts – Use for license bound/compliance applications

EC2 Pricing Cost Models

There are four pricing models for Amazon EC2 instances:

- On-Demand Instances billing are by the second.
- Spot Instances fluctuates periodically based on supply and demand. Supports both per hour and per second (Linux Instances) billing. (Up to 90% Discount)
- Reserved Instances terms are committed 1- or 3-year terms. (Up to 75% Discount)
- Dedicated Hosts are billed the same as On Demand but are billed per hour (Not per second)

EC2 Instances Cost Optimization Best Practices

Use the proper payment approach to minimize costs.

There are three payment options

- All Upfront
- No Upfront
- Partial Upfront

EC2 Instances Cost Optimization Best Practices

EC2 Right Sizing Solutions

- Trusted Advisor C.O checks
- Cost Explorer
- Audits
- Third Party Tools

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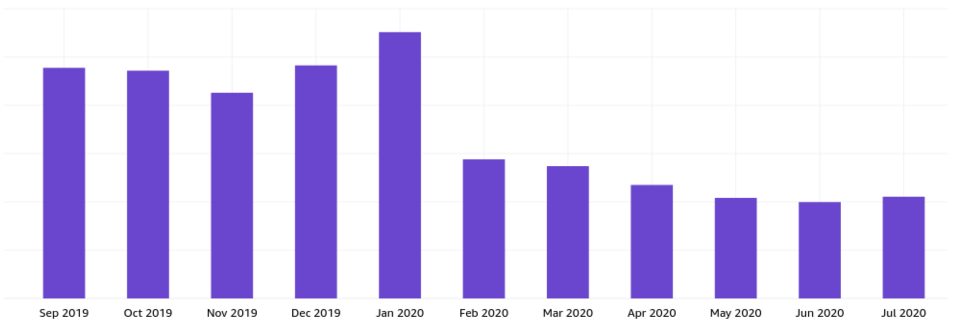
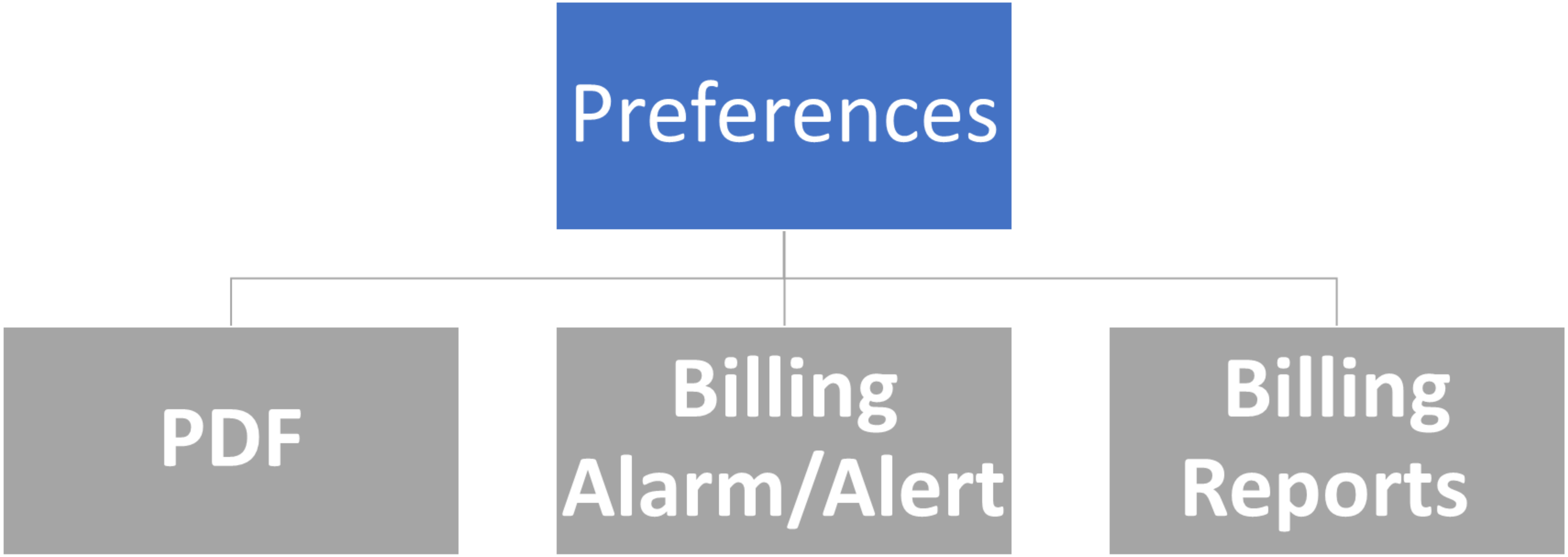


VM Resource Cost Reduction

When deprovisioning a VM ensure your team also removes (deprovisions) these resources that are attached to your VMS.

- Private IP address
- Storage Used

Billing Management



Test Tips

- Billing is per minute initially(Then Per second) and is dependent on several factors.
- If a VM was deployed for 50 seconds, then your charged for a minute.
- Spot instances are good for what type of workload?
- Understand the four EC2 Pricing cost schemes

Demo - AWS Tools

Visibility into your AWS Cloud Spending

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Tools

Whiteboards – AWS FinOps Use Case

Use Case

AWS Use Case

Putting it all Together

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Azure Pricing Model

Managing your Azure Cloud Spending

Azure Charging Variables

Azure VMs are flexible on how you deploy and how pricing is structured

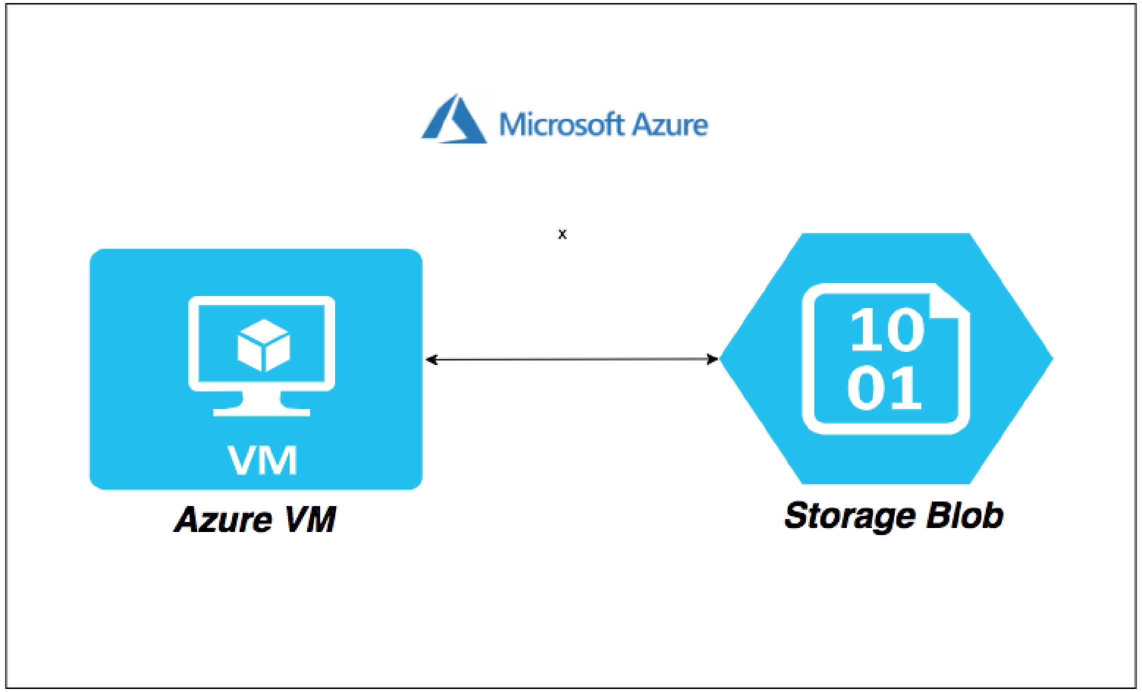
Instances can be priced based on

- Region/Zone
- Machine Image
- Networking
- Time Deployed
- Cost Model

Azure Charging Units

Charging Units

- The minimum unit of time that will be charged is a minute (60 seconds), but after your first minute of time, it is charged for seconds.
- Cost will vary based on region, zone and other factors



Azure Pricing Models

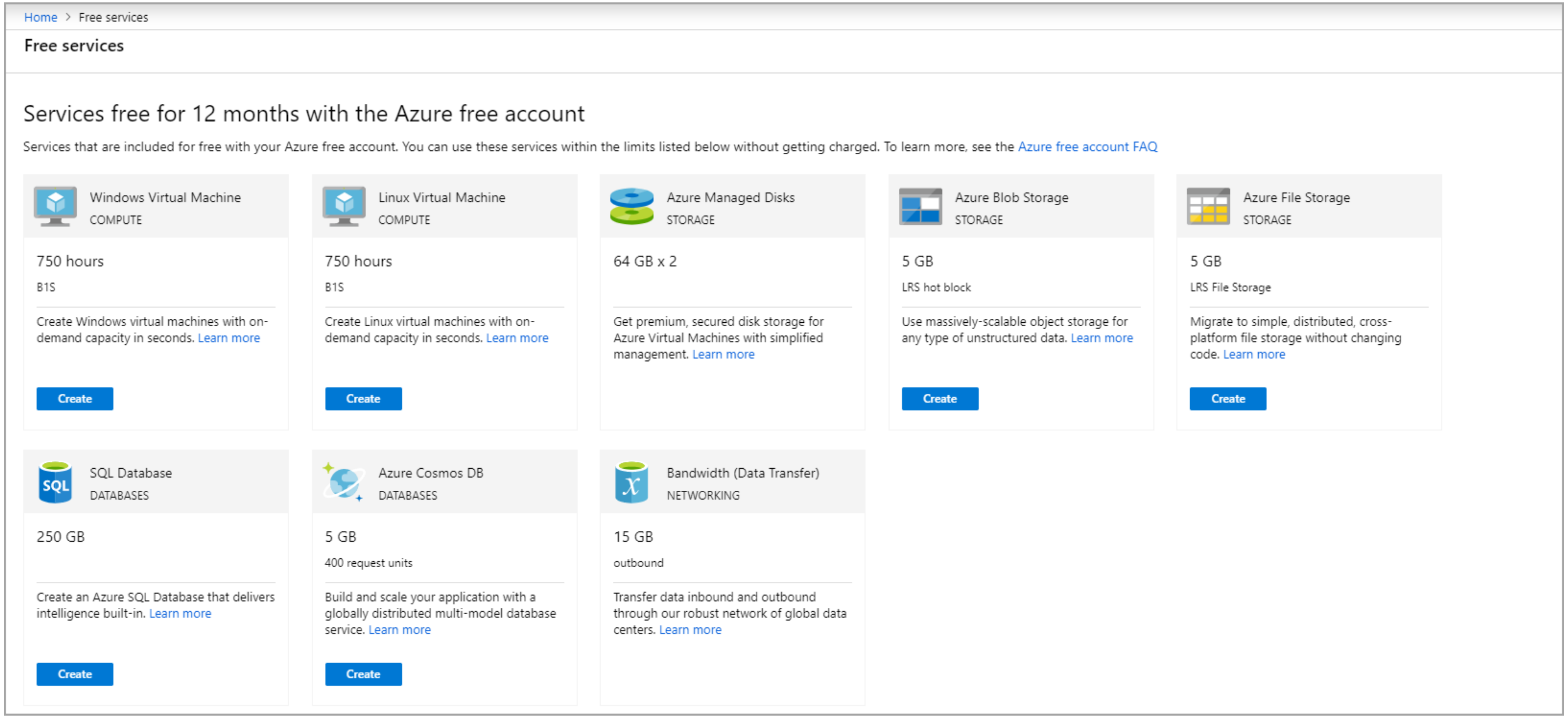
There are three pricing models for Azure VM instances:

- Pay as you Go Instances – Provides for instant use with no commitment
- Reserved Instances – Provides for cost savings with commitment
- Spot Instances – Available Azure capacity at a significant discount for short term

Azure provides several additional options for saving cloud costs.

- Azure Hybrid Benefit program - intended for organizations that own Microsoft licenses in their on-premise data centers.
- Azure service for development and testing - Eligible to substantial discounts.
- Azure pledges to match the cost of equivalent services on AWS - Prices are adjusted every 3 months in line with AWS price changes.

Azure provides Free Services with Account.

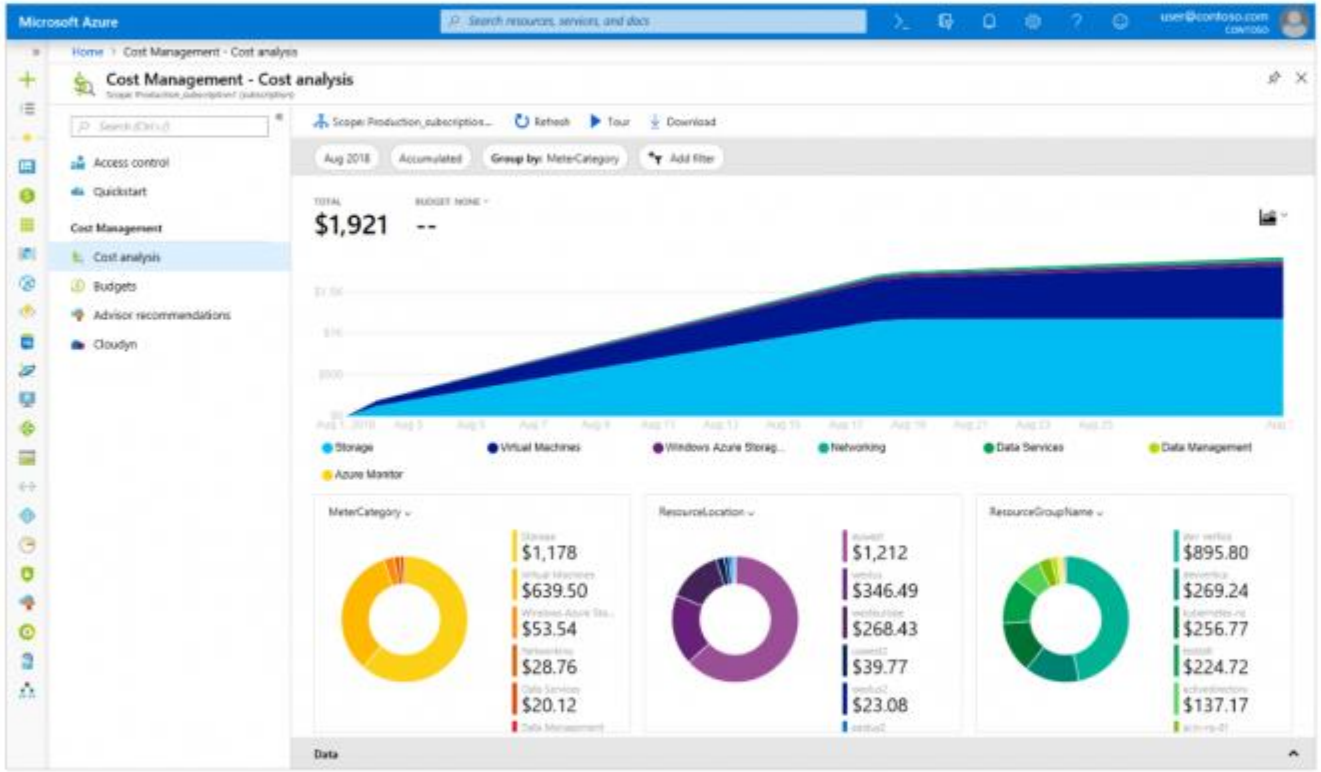


Azure Cost Management Tools

- Azure Price Calculator—can help you estimate the cost of the Azure services you intend to use. You can choose a resource and define the settings. The calculator can then provide you with a cost breakdown.
- Azure Budgets—can help you create a budget that includes costs allocated across the Azure ecosystem. You can set up alert notifications to learn when a certain resource hits a limit or certain thresholds.
- Azure Advisor—can help you gain actionable insights you can use to optimize your costs. This tool analyzes your Azure configurations and uses telemetry to provide you with recommendations to improve availability, security, performance, and costs.
- Azure Exports—can help you export reports generated in Azure. You can use Azure Exports to build custom reports and export the data in a CSV format.

Azure Cost Analysis

- Azure Cost Analysis—can help you get a detailed breakdown of the costs you are spending on Azure resources.
- Provides an in-depth understanding of costs, and filter according to metrics like scope, time, granularity, and types of resources.



Demo – Azure VMs

Provisioning the right VM.

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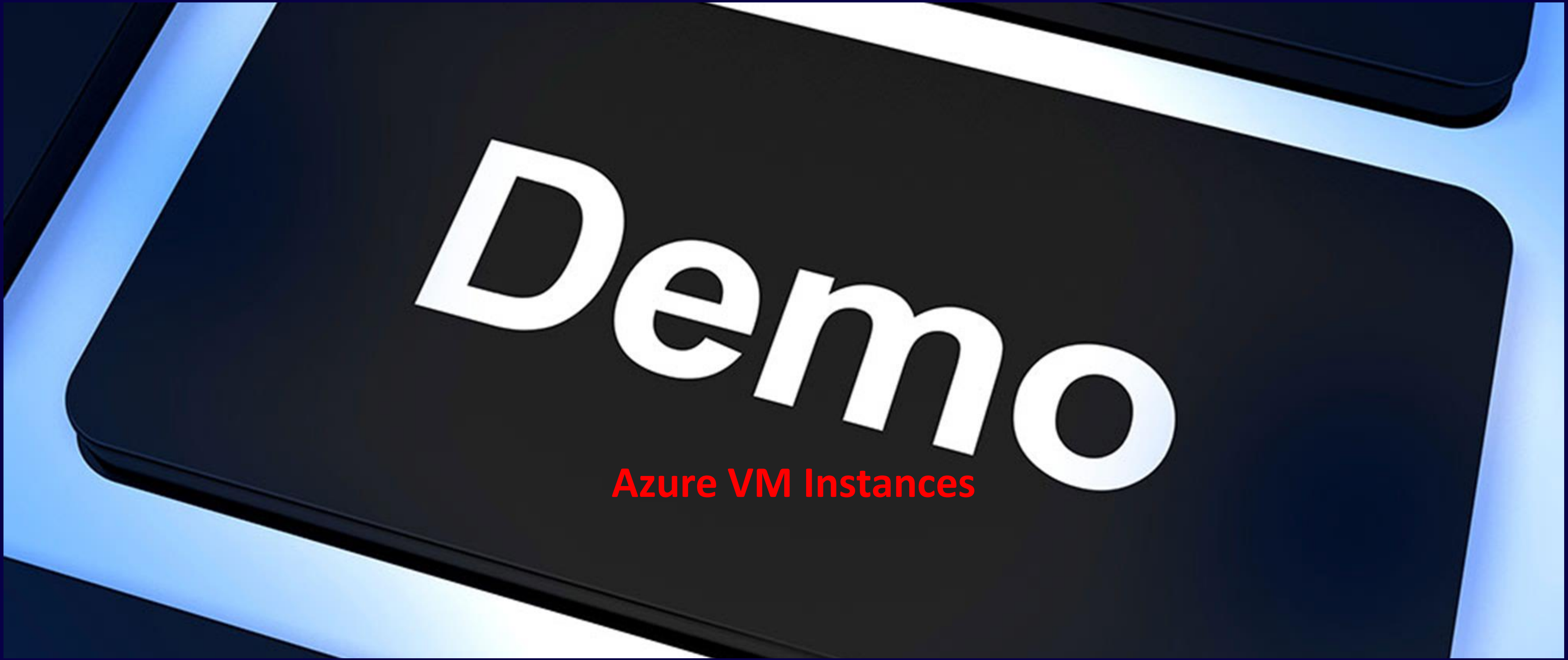


Azure VM Instances

GCP Pricing Model

Managing your GCP Cloud Spending

Google Cloud

- GCP has perhaps the most flexible and transparent model
- GCP has been providing per second billing for years before AWS
- GCP also provides has provided “compute recommendations” automatically with no user intervention.

Billing Discounts

VM vCPU and memory usage for each of these categories can receive one of the following discounts:

- Sustained use discounts
- Committed use discounts
- Discounts for preemptible VM instances

Iowa (us-central1) Monthly Hourly

Item	On-demand price	Preemptible price	1 year commitment price	3 year commitment price
Predefined vCPUs	$0.031611 / vCPU hour	$0.006655 / vCPU hour	$0.019915 / vCPU hour	$0.014225 / vCPU hour
Predefined Memory	$0.004237 / GB hour	$0.000892 / GB hour	$0.002669 / GB hour	$0.001907 / GB hour

https://cloud.google.com/compute/pricing

Discount Model – Sustained Use

- Sustained use discounts are applied on incremental use after you reach certain usage thresholds.
- This means that you pay only for the number of minutes that you use an instance.
- GCP automatically gives you the best price.

Discount Model – Committed Use

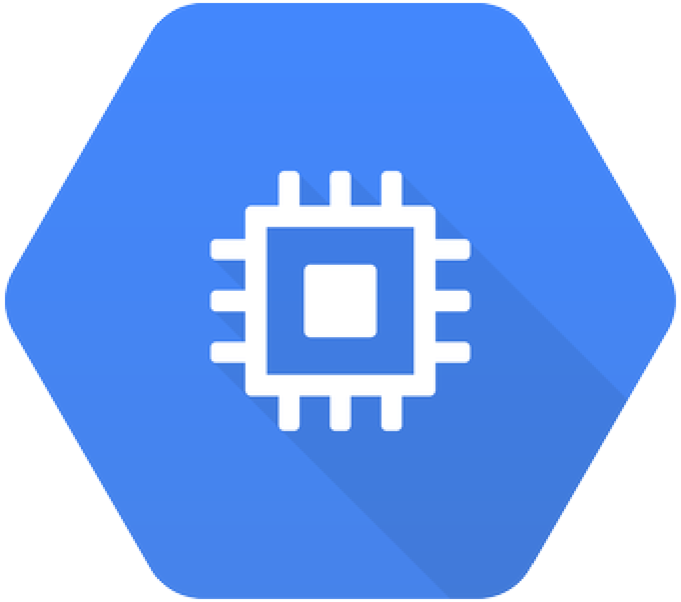
- Committed use discounts are very useful for workloads with predictable resource needs.
- When you purchase a committed use contract, you purchase compute resources at a discounted price.
- Committed use discounts apply to both vCPUs and memory.
- Commitment for 1 year or 3 years

Discount Model – Capacity Reservations

- Reserved capacity is decoupled from reservations
- When you purchase Committed Use Discounts, it's not guaranteed you can launch an instance in the specified region or configuration.
- For a guarantee purchase a zonal reservation.

VM Discounts

- Inferred instances - for billing purposes, the same type of machine used in the same zone will be combined into a single charge (increases your discounts)
- Discount applied to custom machines as well



Billing Model

Per-second billing, sustained use discounts

- 1 minute minimum and 1 sec increments

Preemptible instances

- Live at most 24 hours
- Can be pre-empted with a 30 second notification via API
- Discounted significantly

Test Tips

- Exam had several questions on discount models. Scenario based.
- Charging cycle for self serve accounts are threshold or monthly.
- Committed use discounts are very useful for workloads with predictable resource needs.

Demo – GCP Demo

Visibility into your GCP Spending

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GCP Services

Section Summary

Section 6 - Cloud Resource Pricing and Billing

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6

Section

Section Summary

- Pricing is based on different pricing models, and you should use the proper pricing model for the proper use case.
- The main purpose of Trusted Advisor is to recommend improvements across your AWS account to help optimize and hone your environment based on AWS best practices
- Budgets can be setup to help avoid cost overruns. 20,000 can be setup per account, 2 Free Budgets, Others are at an additional cost.
- Cost Explorer allows gain insight to what your cloud spend has been.
- Azure Cost Analysis—can help you get a detailed breakdown of the costs you are spending on Azure resources.
- VM vCPU and memory usage for each of these categories can receive one of the following discounts - Sustained use discounts, Committed use discounts, Discounts for preemptible VM instances

Section Review Questions

Section 6 – Cloud Resource Planning and Billing

TECH COMMMANDERS
ADVANCING YOUR IT SKILLS

Section Review Questions

Which of the following are the best ways we can manage our billing and determine cost issues? (Select Two)

- Trusted Advisor C.O checks
- Cost Explorer
- Systems Manager
- Contact Support

Section Review Questions

Which of the following are the best ways we can manage our billing and determine cost issues? (Select Two)

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Section Review Questions

Which of the following tool would you want to use to understand previous cloud spending patterns? (Select One)

- Trusted Advisor C.O checks
- Cost Explorer
- Budgets
- Systems Manager

Section Review Questions

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- Trusted Advisor C.O checks
- Cost Explorer
- Budgets
- Systems Manager

Section Review Questions

Which of the following two statements are true about Budgets in AWS? (Select Two)

- Budgets are included with Enterprise and Business support plans
- You receive 2 Free budgets and if you use more than two you will be charged per budget.
- Up to 20,000 can be set up in AWS Budgets
- Budgets can provide insight into past spending habits.

Section Review Questions

Which of the following two statements are true about Budgets in AWS? (Select Two)

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- Up to 20,000 can be set up in AWS Budgets
- Budgets can provide insight into past spending habits.

Section Review Questions

Which of the following statements are correct about charges incurred in AWS? (Select Three)

- There no charges for data transfer inbound (into AWS)
- Data transfer outbound is not charged (out of AWS)
- Storage is paid on a per GB basis
- Compute is paid by the minute or hour
- Support is charged by number of support calls

Section Review Questions

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- Support is charged by number of support calls

Section Review Questions

Your organization is wanting to reserved cloud spending and move off of On Demand instances. What would be benefits of using a Reserved Instance? (Select Three)

- Reserved Instances provide for EC2, RDS, ECS and Lambda savings.
- Reserved Instances provides a “predictable” cost model
- With RIs, you commit in advance for usage which in return means a lower price
- With reservations, you can choose to pay with no-upfront, partial upfront or all upfront

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Section 7 – Course Closeout

Course Wrap-up

Section Overview

EXAM PREPARATION

ADDITIONAL RESOURCES

COURSE CLOSEOUT

Exam Preparation

Let's Discuss

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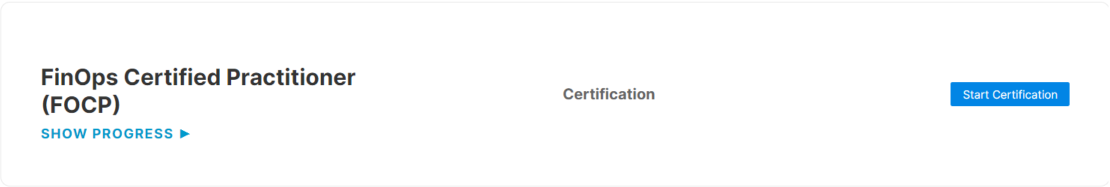
- Let's discuss the exam and how to prepare

- FinOps Exam Registration Affiliate Link http://shrsl.com/3bcox



Additional Resources

Finding additional resources for your FinOps journey

Additional Resources

Resources to learn more FinOps and the exam content

FinOps Certified Practitioner (FOCP) Exam Cheat sheet

- https://www.techcommanders.com/finopspractionercheetsheetpracticequestions
- FinOps Book Available on O'Reilly
- https://learning.oreilly.com/library/view/cloud-finops/9781492054610

Additional Resources

Resources to learn more FinOps and the exam content

FinOps Foundation

- https://www.finops.org/introduction/what-is-finops/

TheGCPGurus

- https://thegcpgurus.com/why-take-the-finops-certified-practitioner-focp-exam-now/

Additional Resources

Resources to learn more FinOps and the exam content

Densify

- https://www.densify.com/finops

EdX

- https://www.edx.org/course/introduction-to-finops

Course Closeout

Thank you for joining

Course Closeout

Thank you for joining the course.

I wish you much success on the exam and in your life!

Please reach out if I can be of assistance.

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- Contact me on LinkedIn
- Contact me on YouTube
- Contact me on Twitter
- Contact me on Steemit
- Contact me at Techcommanders

EMAIL
techie@techcommanders.com

WEBSITE
www.techcommanders.com

PHONE
(904) 512 5529